

Costly State Verification with Limited Commitment

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This paper examines a principal–agent model without transfers in which a Principal mandates actions and conducts costly inspections. The Principal trades off the benefit of mandating her preferred actions against the Agent’s participation, information rents, and inspection costs. I examine which form of commitment matters. Under full commitment, efficient types of the agent are pooled without inspection, intermediate types are inspected with interior probabilities and rewarded for truthful reporting after verification, and inefficient types are excluded. In the limited-commitment cases, the Principal’s preferred equilibrium involves only deterministic inspection. If she commits only to actions, this equilibrium is the optimal full-commitment mechanism restricted to deterministic inspection; if she commits only to inspection, it requires inspecting some inefficient types.

Keywords: Costly state verification, mechanism design, inspection, limited commitment, regulation.

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1 Introduction

Mitigating climate change requires substantial reductions in carbon dioxide emissions. Because firms do not internalize the environmental damages generated by their emissions, unregulated markets generally provide insufficient incentives to abate. Governments therefore rely on a range of policy instruments to induce reductions in emissions and the adoption of cleaner technologies.

Government intervention can take several forms, including emissions taxes, tradable permits, subsidies, support for R&D, and direct regulatory standards. A prominent form of direct regulation is a performance standard, which limits emissions per unit of output. For example, a regulator may impose an upper bound on CO₂ emissions per kilowatt-hour of electricity generated.¹ The appropriate stringency of such a standard may depend on firms' technologies and abatement capabilities, which are often privately known. The regulator does not initially observe the firm's technology but can learn about it by conducting a costly inspection.

In this paper, we study how a regulator's ability to commit to standards and inspection policies separately shapes the optimal regulation. We compare full commitment with environments in which the regulator commits only to standards, only to inspections, or to neither. This comparison allows us to determine the relative value of commitment to each instrument and whether commitment to standards can make inspections credible when the regulator cannot commit to an inspection policy.²

We show that, if the regulator commits to both standards and a stochastic inspection policy, she rewards the Agent by setting a post-inspection standard that

¹See Solomon (2007, Section 13.2.1.1).

²Lack of commitment can stem from several factors which we do not model: First, *legal enforcement*: For legal enforcement to be effective, the court must remain impartial and avoid collusion with any party. Additionally, the contract must be clearly defined and ideally complete. Verifying deviations from the contract can be particularly challenging, especially when it involves non-deterministic promises. Second, *political pressure*: in the context of environmental regulation, regulators may face political pressure by not implement ex-post efficient standards that are ex-ante optimal. For instance, it may be optimal to commit to allowing a firm to pollute more after inspection takes place. Third, *reputation*: reputation can play a significant role in ensuring that the regulator adheres to pre-committed promises, especially when interacting with multiple Agents, either sequentially or simultaneously. Such interactions may enable the regulator to commit to a consistent decision-making process. However, this may require additional assumptions, such as the absence of macroeconomic shocks and the presence of post-facto incentives to align observations with incentive schemes.

departs from the efficient level. The combination of this reward and stochastic inspection reduces information rents. Without commitment to either instrument, this incentive channel is ineffective: although equilibria with stochastic inspection exist in that environment, the regulator prefers an equilibrium relying exclusively on deterministic inspection. When the regulator commits only to standards, her best equilibrium is outcome-equivalent to the optimal mechanism restricted to deterministic inspection. By contrast, when she commits only to inspections, she inspects some inefficient firms, and her maximal equilibrium payoff is lower than the payoff under the deterministic-inspection optimum.

Section 2 introduces a Principal–Agent model in which the Principal (regulator) sets the standards for the Agent (firm). The Agent has a privately known type corresponding to the most stringent standard under which it is willing to participate. Its utility is strictly increasing and strictly concave in the standard: the Agent benefits from a more permissive standard, but its marginal benefit decreases as the permitted pollution level rises. By contrast, for standards above the Agent’s type, the Principal’s utility is strictly decreasing and strictly concave. Thus the efficient standard equals the Agent’s type. The Principal cannot use transfers, but she can conduct a costly inspection that perfectly reveals the Agent’s type. Because the Agent is protected by an ex post participation constraint, he rejects the mandated standard and choose its outside option.³ The Principal trades off the benefit of mandating lower actions against the Agent’s participation, information rents, and inspection costs. Under full commitment, she can commit to the inspection rule and to the standards imposed both with and without inspection; we subsequently relax commitment to each instrument separately.

Section 3 characterizes the optimal mechanism under full commitment. The type space is divided into three ordered, possibly empty regions. Single crossing property implies that the most efficient types are pooled under a common standard and are not inspected. Intermediate types are inspected with strictly interior probabilities. Inspection is needed in this region to limit the gains from mimicking, but deterministic inspection is too costly. In particular, there is no positive-measure region in which inspection occurs with probability one: the Principal can reduce inspection slightly, save the inspection cost, and adjust the standards to preserve incentive compatibility. Finally, the least efficient types are excluded because serving these types is not worth the inspection or information rent.

³Rejection can be interpreted as shutting down, relocating to another jurisdiction, or choosing not to enter the regulated market.

In the interior of the stochastic-inspection region, both standards exceed the Principal’s type-contingent preferred standard, with the post-inspection standard set above the no-inspection standard.⁴ An Agent can benefit from mimicking another type only if the inspection does not occur. The Principal therefore limits the reward provided by the no-inspection standard and shifts a larger share to the post-inspection standard.

Section 4 studies how outcomes change when the Principal can commit to only one of her two regulatory instruments. In Section 4.1, we assume the Principal can commit to standards but cannot commit to an inspection policy. The Principal’s preferred equilibrium is outcome-equivalent to the optimal full-commitment mechanism restricted to deterministic inspection. The key is that commitment to standards allows the Principal to make a deterministic inspection sequentially optimal. Whenever the Agent sends the message that triggers inspection, the Principal commits to a sufficiently stringent standard if she does not inspect. The Agent rejects this standard, so a deviation by the Principal from the prescribed inspection decision yields her no surplus. By committing to this self-punishing fall-back, the Principal makes it optimal to inspect whenever inspection is prescribed, even though she cannot commit directly to doing so. The Principal cannot improve on the deterministic benchmark through stochastic inspection. Randomization can occur only when she is indifferent between inspecting and not inspecting after a message. Such randomization can be replaced by deterministic inspection without reducing her payoff, while full inspection weakly reduces the information rents needed to sustain truthful reporting. The Principal’s best equilibrium when she commits only to standards is therefore equal in value to the optimal deterministic full-commitment mechanism.

Section 4.2 considers the reverse case: the Principal commits to an inspection policy but cannot commit to standards. After inspection, she observes the Agent’s type and imposes her ex post preferred standard. Without inspection, she chooses the standard that is optimal given her posterior belief. Although equilibria with stochastic inspection exist, it is sufficient to focus on deterministic inspection when searching for the Principal’s preferred equilibrium. The equilibrium has the same structure as the optimal deterministic-inspection mechanism, with one exception: the Principal must inspect, rather than exclude, some inefficient types. Inspecting these types makes it optimal for her to exclude the remaining types. Consequently, the Principal’s payoff is weakly lower than under commitment

⁴Note that if the Agent misreports its type, inspection reveals its true type, and the Principal excludes it; this contingency does not arise on the equilibrium path.

only to standards, which delivers the deterministic full-commitment value. If the optimal deterministic full-commitment mechanism has no exclusion region, however, the Principal’s preferred equilibria under commitment only to inspections and commitment only to standards are outcome-equivalent.

Finally, we consider two extensions. Section 5.1 analyzes the case in which the Principal has no commitment power. Section 5.2 considers partial commitment, under which the Principal can commit to inspect whenever the Agent requests an inspection. If the optimal deterministic benchmark has no exclusion region, this form of limited commitment implements the same deterministic full-commitment benchmark.

Relationship to the literature. This paper contributes to the literature on costly state verification (CSV). The classical CSV framework begins with [Townsend \(1979\)](#), who studies an optimal insurance problem between an investor and an entrepreneur. After the contract is signed, the entrepreneur privately observes the project’s realized income and reports it. Verification of the report is costly, and the optimal contract specifies when verification occurs and what transfer is made. The resulting contract resembles a debt contract: verification occurs only for low reported income. [Gale and Hellwig \(1985\)](#) study a related borrower–lender problem with risk-neutral parties and show that the optimal contract again has a debt-like structure. These foundational papers focus on financial contracts with transfers and deterministic verification.

Subsequent work allows for stochastic verification. [Border and Sobel \(1987\)](#) study a more general mechanism with stochastic inspections and bounded pre-inspection and post-inspection transfers, under the restriction that the Principal cannot make a net payment to the Agent. They show that inspection probabilities may decrease with the Agent’s wealth and provide examples in which optimal contracts use large rewards together with infrequent inspections. [Mookherjee and Png \(1989\)](#) show that stochastic verification can also arise when the borrower is risk averse. These papers are closely related to our full-commitment benchmark, where stochastic inspection can be useful because the Principal can reward truthful reports after inspection. A central difference is that our model does not use transfers, and the Agent is protected by an ex-post participation. This changes both the form of rewards and the scope for punishment.⁵

⁵There are also other differences. Many canonical financial CSV models study credit or insurance contracts and maximize the borrower’s or entrepreneur’s expected utility subject to the lender’s or investor’s participation, break-even, or zero-profit constraint. In addition, the hidden information is typically realized income or output in a transfer contract, whereas in our

Our work also relates to costly verification in allocation and collective-choice environments without transfers. [Ben-Porath et al. \(2014\)](#) study a Principal who allocates an object to one of several Agents and can verify their private information at a cost. [Mylovanov and Zapechelnyuk \(2017\)](#) study a related allocation problem with a different verification technology and limited punishments, in which verification can occur after allocation and the Principal can destroy part of the Agent’s payoff. [Li \(2020\)](#) analyzes the connection between costly verification and limited punishment. [Patel and Urgun \(2022\)](#) introduce money burning as an additional instrument, while [Erlanson and Kleiner \(2020\)](#) study optimal allocation and collective choice with costly verification. These papers share our focus on using costly verification to deter misreporting when transfers are unavailable or punishments are limited, but they do not study how the optimal mechanism varies with the Principal’s commitment power.

Another related branch studies costly verification in monopoly regulation. [Baron and Besanko \(1984\)](#) extend [Baron and Myerson \(1982\)](#) by allowing for random and costly inspections in a regulation model with two-part tariffs. [Palonen and Pekkarinen \(2022\)](#) consider a regulatory CSV problem in which the Agent can undertake costly avoidance to reduce the probability of verification. In their model, punishment following a detected misreport is exogenous and linear, and truthful reports are not rewarded following verification. Unlike our model, theirs also allows monetary transfers. Moreover, our model differs further in two ways. First, the Agent has an ex post participation right, so punishments are endogenous and bounded by the Agent’s option to reject. Second, our main focus is the comparison across commitment regimes.

The paper most closely related to ours is [Halac and Yared \(2020\)](#), who study a costly-verification delegation problem in which the Agent is biased toward higher actions. When the Agent’s bias is large and the Principal has full commitment, their environment is close in spirit to our benchmark. The main differences are that we allow stochastic mechanisms, impose an ex post participation constraint, and study how the optimal mechanism changes as the Principal’s commitment power is weakened. In particular, our full-commitment analysis shows how stochastic inspection can be used to deliver rewards after truthful verification, while our limited-commitment results show why such rewards may fail to be credible once the Principal cannot commit to ex-post inefficient actions.⁶

model the private information is a type that determines the appropriate real action, and the Agent is protected by ex-post participation.

⁶Further differences are discussed when we introduce the deterministic inspection benchmark

The limited-commitment and no-commitment parts of the paper are also connected to the cheap-talk literature following [Crawford and Sobel \(1982\)](#). When the Principal cannot commit, messages become partly cheap talk: the Agent reports private information, and the Principal responds after observing the message. Our environment is not a standard cheap-talk model, however, because the Principal may verify the report at a cost and the Agent can accept or reject the final mandated action.⁷ [Khalil \(1997\)](#) studies costly auditing without commitment to an inspection policy, but his model relies on costly signaling rather than cheap talk. [Melumad and Mookherjee \(1989\)](#) also studies a setting without commitment to inspection; unlike in our model, the Principal can commit to transfers after inspection and the Agent cannot reject the mechanism *ex post*.

Finally, [Banks \(1989\)](#) studies a model in which an Agent requests a budget and the Principal lacks commitment power. Utilities are linear in transfers and inspection is imperfect. In their setting, full pooling is the only equilibrium, so the Agent transmits no information to the Principal. The contrast with our analysis highlights the role of perfect verification and commitment. With perfect inspection, a misreport can be detected and punished, which allows a semi-separating equilibrium to be sustained.

2 Model

Players and information structure. There are two players, a Principal (she) and an Agent (he). The Agent has a private type $\theta \in \Theta = [\underline{\theta}, \bar{\theta}]$ drawn from a commonly known cumulative distribution function $F(\cdot)$ with density $f(\cdot) > 0$.

Strategies and actions. The Agent sends a message m from $M = [0, 1]$. Let $\sigma(\cdot | \theta) \in \Delta(M)$ denote the probability distribution over messages chosen by an Agent of type θ , where $\Delta(M)$ is the set of all probability measures over M .⁸ Upon receiving m , the Principal inspects or not. Let $i(m) \in [0, 1]$ denote the probability that the Principal conducts an inspection. Inspection allows the Principal to learn the Agent's type θ . Contingent on the inspection decision, the Principal mandates an action: if no inspection occurs, she mandates $a(m) \in \mathbb{R}_+$; if an inspection

and the commitment-only-to-inspection environment in Proposition 3.

⁷Thus the sender's message is cheap in the usual sense, but the receiver's response is constrained by costly verification and by the Agent's *ex-post* participation right.

⁸We require σ to be a stochastic kernel from $(\Theta, \mathcal{B}(\Theta))$ to $(M, \mathcal{B}(M))$: for every Borel set $B \subseteq M$, the map $\theta \mapsto \sigma(B | \theta)$ is Borel measurable.

occurs, she mandates $a^I(m, \theta) \in \mathbb{R}_+$.

Payoffs. Inspection entails a cost $\phi > 0$ to the Principal. The Agent is protected by **ex-post participation**: he can reject the mandated action. If the Agent rejects the action, both players receive a payoff of zero. The payoff of the Agent with type θ and action a is denoted $u(\theta, a)$.⁹ The Principal's payoff is given by

$$v(\theta, a) \mathbb{1}_{\text{accept}} - \phi \mathbb{1}_{\text{inspection}}.$$

Specifically, we assume:

$$u(\theta, a) = (a - \theta + 1)^\alpha - 1,$$

$$v(\theta, a) = v_0(\theta) - (a - \theta)^\beta,$$

where $v_0(\theta) > 0$ is weakly decreasing in θ , and $0 < \alpha < 1 < \beta$.^{10,11}

Timing. The sequence of events is as follows: Nature draws a type θ , which the Agent observes privately. The Agent then sends a message $m \in M$. The Principal observes m and inspects the Agent with probability $i(m)$. Depending on the inspection outcome, she mandates either $a(m)$ or $a^I(m, \theta)$. Finally, the Agent accepts or rejects the mandated action. Figure 2 illustrates this sequence.

For each commitment regime—commitment to actions, to inspection, or to both—we first fix the relevant policy instruments and characterize the Principal's preferred equilibrium in the resulting continuation game.¹² We then optimize over the instruments to which the Principal can commit and identify the policy that maximizes her ex ante payoff.¹³

In the context of environmental regulation, the mandated action a represents the permitted pollution level (or standard) granted to the firm. The state variable

⁹An alternative interpretation is that the Agent possesses a private outside option dependent on θ .

¹⁰The Agent's utility incorporates a standard normalization. The constant 1 ensures that the marginal utility remains well-defined and finite when the action equals the firm's true type ($a = \theta$), avoiding division by zero in the derivative. Correspondingly, subtracting 1 outside the power term normalizes the Agent's baseline utility such that mandating the efficient action yields a reservation payoff of exactly zero, $u(\theta, \theta) = 0$.

¹¹In Appendix 6, we consider a broader class of utility functions, formalized in Assumption 1.

¹²We use Perfect Bayesian Equilibrium and impose the belief-consistency requirements of Fudenberg and Tirole (1991).

¹³Of course, under full-commitment, this is no longer a game, as instruments of the principal are fixed.

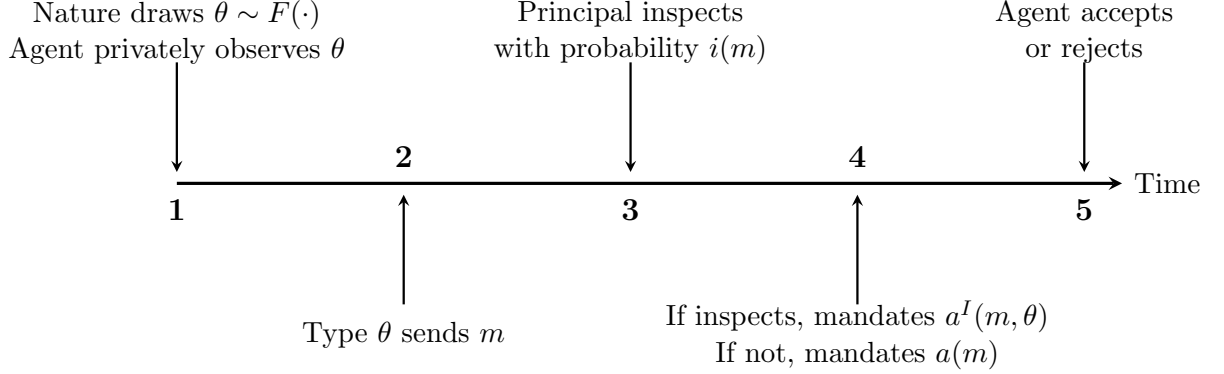


Figure 1: Timing of the game.

θ represents the firm's private technological type or baseline abatement capability. A "clean" firm (low θ) can achieve low emissions efficiently, whereas a "dirty" firm (high θ) finds it prohibitively expensive to operate at low emission levels. Consequently, θ represents the regulator's preferred, ex-post efficient pollution target for that specific firm.

The payoff functions capture the fundamental economic trade-offs of environmental policy. The Principal's utility features a strictly convex loss function, representing strictly convex environmental damages. This reflects the ecological reality that while initial units of pollution may be safely absorbed by the environment, marginal damages accelerate severely as emissions increase (e.g., atmospheric saturation leading to exponential health and environmental costs).¹⁴

Conversely, the Agent's utility is strictly concave in a . Because the firm's utility is derived from the avoidance of abatement costs, this captures diminishing marginal cost savings. When regulations are highly stringent (low a), granting one additional unit of permitted pollution saves the firm substantial costs. However, if the firm is already polluting heavily, an additional unit yields negligible cost savings.¹⁵

Pairing strictly convex damages ($\beta > 1$) with strictly concave cost savings ($\alpha < 1$) ensures that the marginal rate of substitution diverges ($\lim_{a \rightarrow \infty} -v_a/u_a = \infty$).

¹⁴Convex loss functions (such as the canonical quadratic form) for the regulator are the foundational standard in both environmental economics and the broader strategic communication literature. See Weitzman (1974) for the classic application to emission quotas, and Crawford & Sobel (1982) for the general theoretical framework of biased policy bliss points.

¹⁵The assumption of strictly convex abatement costs (and therefore strictly concave utility from pollution allowances) is standard in the mechanism design approach to regulation; see, for instance, Laffont & Tirole (1993).

This strictly increasing marginal cost reflects that at extreme pollution levels, environmental destruction vastly outpaces any private economic benefit to the firm. This boundary condition guarantees the existence of optimal regulatory contracts when the action space is unbounded.

We explore environments where the Principal possesses varying degrees of commitment ability and how these variations affect the optimal regulatory policies.

3 Full Commitment

Suppose the Principal fully commits to an inspection policy and mandated actions. Under full commitment, the Principal designs and publicly commits to a mechanism before the Agent observes his private type. By the revelation principle, we restrict attention to direct truthful mechanisms,

$$\mathbb{M} = (i(\hat{\theta}), a(\hat{\theta}), a^I(\hat{\theta}, \theta)).$$

The Agent reports a type $\hat{\theta} \in [\underline{\theta}, \bar{\theta}]$. The Principal then inspects the Agent with probability $i(\hat{\theta})$. If no inspection occurs, she mandates the action $a(\hat{\theta})$. If inspection occurs, she perfectly observes the true type θ and mandates the post-inspection action $a^I(\hat{\theta}, \theta)$.

Under full commitment, any detected misreport is punished maximally. In particular, whenever $\hat{\theta} \neq \theta$, the Principal sets

$$a^I(\hat{\theta}, \theta) \leq \underline{\theta},$$

which the Agent rejects ex post whenever it lies below his type, yielding him zero payoff. Thus, after a detected misreport, the Agent obtains no payoff. The mechanism-design problem is therefore to choose the inspection probabilities $i(\theta)$, the uninspected actions $a(\theta)$, and the truthful post-inspection actions $a^I(\theta, \theta)$.

Incentive compatibility requires every type $\theta \in \Theta$ to weakly prefer reporting truthfully to reporting any $\hat{\theta} \in \Theta$:

$$\begin{aligned} & (1 - i(\theta))u(\theta, a(\theta))\mathbb{1}_{\{a(\theta) \geq \theta\}} + i(\theta)u(\theta, a^I(\theta, \theta))\mathbb{1}_{\{a^I(\theta, \theta) \geq \theta\}} \\ & \geq (1 - i(\hat{\theta}))u(\theta, a(\hat{\theta}))\mathbb{1}_{\{a(\hat{\theta}) \geq \theta\}} + i(\hat{\theta})u(\theta, a^I(\hat{\theta}, \theta))\mathbb{1}_{\{a^I(\hat{\theta}, \theta) \geq \theta\}}, \quad \forall \theta, \hat{\theta} \in \Theta. \end{aligned}$$

Under maximal punishment, a detected misreport is assigned a standard that

the Agent rejects. The inspection branch of any deviation therefore yields zero utility, and the incentive-compatibility condition reduces to

$$\begin{aligned} & (1 - i(\theta))u(\theta, a(\theta))\mathbb{1}_{\{a(\theta) \geq \theta\}} + i(\theta)u(\theta, a^I(\theta, \theta))\mathbb{1}_{\{a^I(\theta, \theta) \geq \theta\}} \\ & \geq \sup_{\hat{\theta} \in \Theta} (1 - i(\hat{\theta}))u(\theta, a(\hat{\theta}))\mathbb{1}_{\{a(\hat{\theta}) \geq \theta\}}, \quad \forall \theta \in \Theta. \end{aligned} \tag{IC}$$

The Principal chooses the inspection rule and the standards imposed with and without inspection to maximize her ex ante payoff subject to (IC) and feasibility.¹⁶

The first proposition characterizes the structure of the optimal full-commitment mechanism. The key implication is that the optimal full-commitment mechanism contains, up to null sets, three possible regions: a no-inspection pooling region, a stochastic inspection region, and an exclusion region. Moreover, a deterministic full-inspection region cannot have positive measure.

Proposition 1 *The optimal mechanism is unique¹⁷, and*

$$\mathbb{M} = (i(\cdot), a(\cdot), a^I(\cdot, \cdot))$$

is characterized, up to null sets, by two thresholds

$$\underline{\theta} \leq \theta_c \leq \theta_e \leq \bar{\theta}.$$

The mechanism has the following structure:

(i) **No-Inspection Region** ($\theta < \theta_c$): *The Principal does not inspect,*

$$i(\theta) = 0,$$

and mandates a constant uninspected action

$$a(\theta) = a_c \geq \theta_c.$$

(ii) **Stochastic Inspection Region** ($\theta \in [\theta_c, \theta_e]$): *The Principal conducts*

¹⁶Formally, the Principal solves

$$\begin{aligned} & \max_{i(\cdot), a(\cdot), a^I(\cdot, \cdot)} \int_{\underline{\theta}}^{\bar{\theta}} \left[(1 - i(\theta))v(\theta, a(\theta))\mathbb{1}_{\{a(\theta) \geq \theta\}} + i(\theta) \left(v(\theta, a^I(\theta, \theta))\mathbb{1}_{\{a^I(\theta, \theta) \geq \theta\}} - \phi \right) \right] f(\theta) d\theta \\ & \text{s.t. (IC),} \quad i : \Theta \rightarrow [0, 1], \quad a : \Theta \rightarrow \mathbb{R}_+, \quad a^I : \Theta^2 \rightarrow \mathbb{R}_+. \end{aligned}$$

¹⁷Uniqueness is understood up to outcome-equivalent choices of actions that the Agent rejects.

randomized inspections,

$$0 < i(\theta) < 1.$$

In this region,

$$a^I(\theta, \theta) > a(\theta) > \theta.$$

(iii) **Exclusion Region** ($\theta > \theta_e$): The Principal does not inspect,

$$i(\theta) = 0,$$

and mandates the action $a(\theta) = \underline{\theta}$ which the Agent rejects.

Figure 2 illustrates the action rules $a(\theta)$ and $a^I(\theta, \theta)$, while Figure 3 illustrates the inspection rule $i(\theta)$.¹⁸

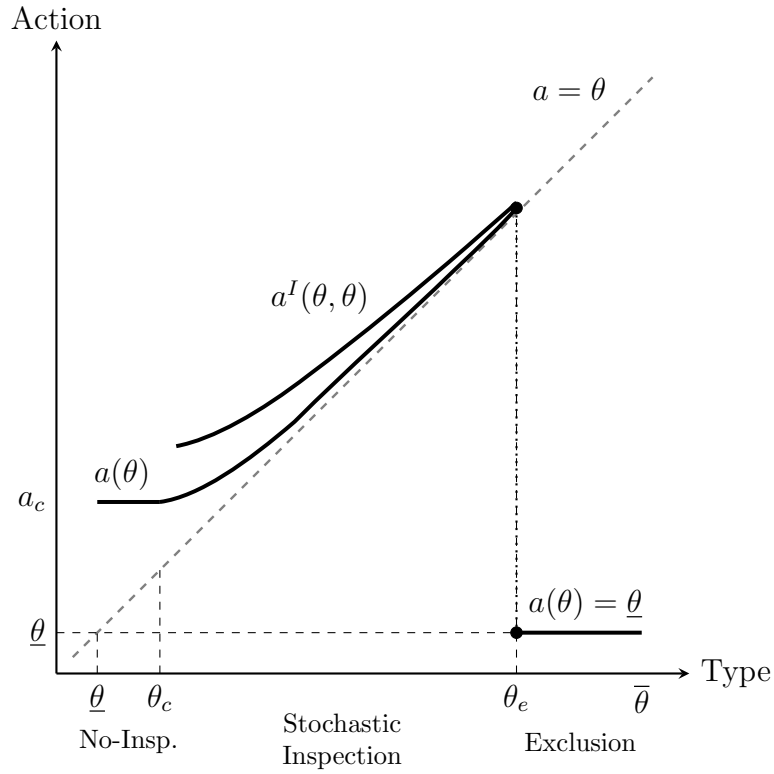


Figure 2: Mandated actions in the optimal full-commitment mechanism.

¹⁸Figures 2 and 3 are obtained from a numerical solution of the optimal full-commitment problem.

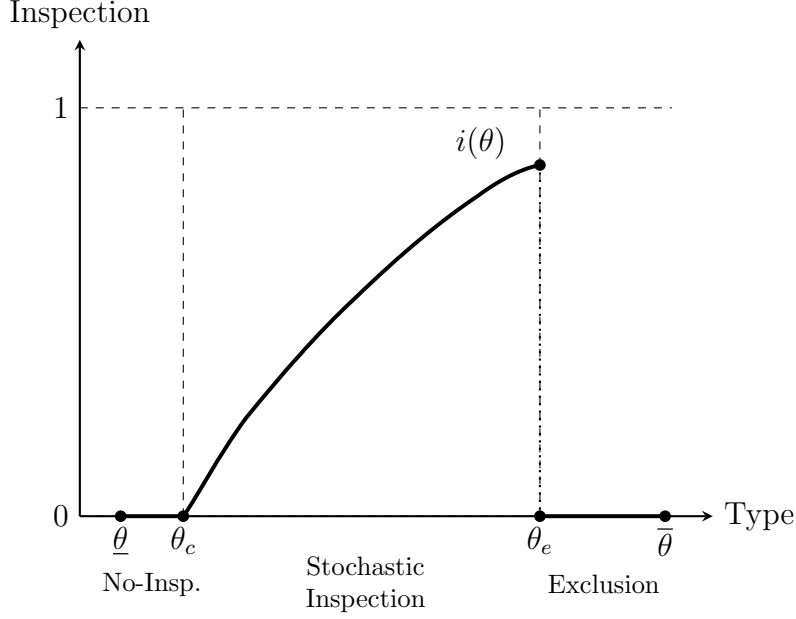


Figure 3: Inspection in the optimal full-commitment mechanism.

No-Inspection Region: The proposition states that the no-inspection region is a pooling region. If two non-excluded types $\theta_1 < \theta_2$ are both not inspected, then type θ_1 can accept the action assigned to θ_2 . Incentive compatibility and $u_a > 0$ imply $a(\theta_1) \geq a(\theta_2)$. But then type θ_2 can also accept the action assigned to θ_1 , and incentive compatibility gives the reverse inequality. Hence all non-excluded zero-inspection types must receive the same action, denoted a_c .

This pooling region lies at the bottom of the type space and cannot appear after a positive-inspection region. If a type θ_0 is served without inspection at action a_c , then any lower type $t < \theta_0$ can also be assigned the same action without creating a new profitable deviation for type t . To see this, consider any report $\hat{\theta}$. If $a(\hat{\theta}) \leq a_c$, then $u_a > 0$ implies that type t weakly prefers a_c to $a(\hat{\theta})$. If instead $a(\hat{\theta}) > a_c$, log-supermodularity of u implies that the higher action is relatively more attractive to θ_0 than to t . Hence, if type t wants to mimic report $\hat{\theta}$ rather than take a_c , then type θ_0 would also want to mimic report $\hat{\theta}$, contradicting incentive compatibility of θ_0 . Assigning type t to the same no-inspection action a_c is also weakly better for the Principal. Since type t could already mimic θ_0 , the original mechanism must give type t at least the utility $u(t, a_c)$. For fixed t , the marginal cost of delivering

utility through action a is

$$MC(t, a) = -\frac{v_a(t, a)}{u_a(t, a)}.$$

Because $MC(t, a)$ is increasing in the action a , the Principal's payoff as a function of the utility promised to type t is decreasing and concave.¹⁹ Thus, delivering the required utility through a_c is weakly cheaper than delivering at least the same utility through a lottery over $a(t)$ and $a^I(t, t)$. Inspection costs make the original allocation even less attractive. Therefore any lower type who is inspected can be moved to the same no-inspection cap, which is why a zero-inspection region cannot appear after a positive-inspection region.

Stochastic Inspection Region: In the stochastic inspection region, the action mandated after inspection and truthful reporting must exceed the uninspected action, and the uninspected action must itself exceed the efficient action:

$$a^I(\theta, \theta) > a(\theta) > \theta.$$

Intuitively, the Principal uses the inspected branch to reward a truthful Agent: if inspection takes place and the report is verified, the Agent receives the larger action $a^I(\theta, \theta)$.

To see why $a^I(\theta, \theta) > a(\theta)$, first suppose the reverse is true. Consider lowering $a(\theta)$ and raising $a^I(\theta, \theta)$, while keeping the expected payoff of type θ fixed. This perturbation weakly relaxes incentive constraints because it lowers the no-inspection action, which is the only branch that mimics can benefit from. The Principal also benefits from this perturbation. Holding the inspection probability fixed, the perturbation is a mean-preserving contraction of type θ 's utility. The Principal's payoff as a function of the Agent's utility is strictly concave. Therefore this mean-preserving contraction strictly increases the Principal's payoff. Hence $a(\theta) > a^I(\theta, \theta)$ cannot be optimal.

If $a^I(\theta, \theta) = a(\theta)$, then inspection does not change the action; it only reduces mimicking payoffs because detected mimics are punished. The Principal can reduce the inspection probability slightly, lower the uninspected action enough so that the deviation payoff from report θ does not increase, and raise the inspected action to

¹⁹For fixed type t , let $a_t(r)$ solve $u(t, a_t(r)) = r$ and define $\Psi_t(r) = v(t, a_t(r))$. Then $\Psi'_t(r) = v_a(t, a_t(r))/u_a(t, a_t(r)) = -MC(t, a_t(r))$. Since $u_a > 0$, $a_t(r)$ is increasing in r . Thus, if $MC(t, a)$ is increasing in a , then $MC(t, a_t(r))$ is increasing in r , so $\Psi'_t(r)$ is decreasing. Hence Ψ_t is concave. Since $MC(t, a) > 0$, $\Psi'_t(r) < 0$, so Ψ_t is decreasing.

keep type θ 's payoff fixed. The Principal benefits from this variation. Since the two actions initially coincide, the action adjustment has only a second-order effect while the saving in inspection cost is first order. Therefore $a^I(\theta, \theta) > a(\theta)$.

The no-inspection action in the stochastic region must be above the Principal's preferred action: $a(\theta) > \theta$. To see the intuition, suppose instead that $a(\theta) < \theta$. Then type θ rejects the no-inspection action. Hence, without loss, the Principal can exclude this type upon no inspection, $a(\theta) = \underline{\theta}$, because this branch gives zero payoff to both players anyway. Now, if $v(\theta, a^I(\theta, \theta)) - \phi \geq 0$, both the Principal and type θ receive a higher payoff if the Principal inspects with probability one. Therefore, the choice of a is irrelevant. If instead $v(\theta, a^I(\theta, \theta)) - \phi < 0$, then the Principal loses utility whenever type θ is inspected, and she also receives zero when type θ is not inspected because the type rejects the no-inspection action. Thus the expected payoff from serving type θ is negative. Intuitively, this suggests that types with rejected no-inspection branches and loss-making inspected branches should be excluded rather than served in the stochastic region. Hence $a(\theta) > \theta$.

No Full Inspection Region: Suppose the Principal inspects type θ , which has a strictly positive rent. Since types above θ_e receive zero rent, no no-inspection action can exceed θ_e and therefore $a^I(\theta, \theta) \leq \theta_e$. Now take a positive-measure subset of such fully inspected types and choose $\rho > 0$ so that, on this subset, $\theta < a^I(\theta, \theta) - \rho \leq \theta_e - \rho$. The Principal can replace full inspection by almost-full inspection as follows. With small probability ε , she does not inspect and assigns $a^I(\theta, \theta) - \rho$. With probability $1 - \varepsilon$, she inspects and assigns a slightly higher action $\tilde{a}^I(\theta, \theta)$ chosen to keep type θ 's payoff unchanged. The gain from this perturbation is the reduction in inspection cost. The loss is the action adjustment needed to keep the type's payoff unchanged: the Agent receives the slightly lower no-inspection action $a^I(\theta, \theta) - \rho$, and this must be offset by a slightly higher inspected action. This loss can be made arbitrarily small by choosing ρ small. In contrast, the inspection-cost saving does not disappear as ρ becomes small: each unit of reduced inspection saves the fixed cost ϕ . Therefore the Principal strictly gains.

The perturbation also preserves incentive compatibility. The new no-inspection action is below $\theta_e - \rho$, so types above $\theta_e - \rho$ cannot accept it and obtain positive payoff. Types below $\theta_e - \rho$ already have positive rents, bounded away from zero, while the new deviation payoff created by this perturbation is only of order ε . Hence, for small enough ε , no new deviation is profitable. Thus full inspection cannot occur on a positive-measure subset of the stochastic region, and the middle region satisfies $0 < i(\theta) < 1$.

A full-inspection region cannot exist after rents have fallen to zero. Suppose a positive mass of zero-rent types is inspected with probability one. This set must lie at the beginning of the exclusion region. Otherwise, the Principal could inspect some lower types that are currently excluded and instead exclude an equal mass of higher types that are currently fully inspected, thereby increasing her payoff, since $v(\theta, \theta)$ is decreasing in θ . Because these types receive zero payoff, each of them must be assigned the action θ after inspection. The Principal can reduce their inspection probability slightly and assign the same action θ whenever inspection does not occur. This leaves each type's payoff unchanged, since $u(\theta, \theta) = 0$, but it saves the inspection cost on a positive-measure set. Hence the inspection-cost saving is of order ε .

This perturbation may induce lower zero-rent types to mimic. Their new deviation payoff is also proportional to ε , so they can be compensated by raising their post inspection action from t to $t + d_\varepsilon(t)$. Since $u_a(t, t) > 0$, delivering utility of order ε requires an action increase of the same order. However,

$$0 \leq v(t, t) - v(t, t + d_\varepsilon(t)) \leq K d_\varepsilon(t)^\gamma, \quad \gamma > 1.$$

Hence the Principal's loss is higher order of ε .

Exclusion Region: The Principal excludes the most inefficient types. Once types no longer receive positive rents, serving them without creating new mimicking incentives becomes costly. If a type $\theta > \theta_e$ is assigned the uninspected action $a(\theta) = \theta$, then types in (θ_e, θ) would want to mimic θ unless inspection is full, which has already been ruled out. Moreover, if inspection is not deterministic, $a^I(\theta, \theta) = \theta$, and $a(\theta) < \theta$ then the Principal would either want to inspect this type with probability one when $v(\theta, \theta) - \phi > 0$, or not inspect and exclude the type when $v(\theta, \theta) - \phi < 0$. The first case cannot occur on a positive-measure set because full inspection has already been shown to be suboptimal. Therefore types above θ_e are excluded. Exclusion is implemented by setting

$$i(\theta) = 0, \quad a(\theta) = \underline{\theta},$$

so that the Agent rejects and both players receive zero. This is why the action rule drops to the rejected action in Figure 2, and the inspection probability drops to zero in Figure 3.

Existence of the Optimal Mechanism: The existence of an optimal full-commitment mechanism is not automatic because the action space is unbounded.

The key force guaranteeing existence is that large rewards eventually become prohibitively costly for the Principal. Holding the Agent’s promised utility fixed, increasing $a^I(\theta, \theta)$ allows the Principal to reduce the inspection probability and save inspection costs. However, the inspection-cost saving is bounded by the fixed cost ϕ , whereas the marginal cost of delivering utility through the inspected reward,

$$MC(\theta, a^I) = -\frac{v_a(\theta, a^I)}{u_a(\theta, a^I)},$$

becomes arbitrarily large as $a^I \rightarrow \infty$. Therefore, for sufficiently large inspected rewards, the additional loss in the Principal’s action payoff overwhelms the inspection-cost saving, and the Principal’s net marginal payoff from increasing $a^I(\theta, \theta)$ becomes arbitrarily negative. Hence the optimal inspected reward must be bounded.²⁰

Implications for Environmental Regulation. In the optimal mechanism whenever $a^I(\theta, \theta)$ is relevant, $a^I(\theta, \theta) > a(\theta) > \theta$. In the context environmental regulation, this means that after verifying a firm’s type, the regulator may commit to granting the truthful firm a pollution allowance above its ex-post efficient standard. This is precisely the kind of ex-post inefficient reward that full commitment permits. The result highlights why full commitment is a useful benchmark but an unrealistic institutional assumption: once the regulator observes the true type, it may be difficult to credibly implement a pollution standard that is knowingly too lax. This motivates the limited-commitment environments studied below.²¹

4 Different Commitment Ability

In this section, we relax the assumption of full commitment to examine environments where the Principal possesses limited commitment power over her regulatory instruments. Specifically, we analyze two polar cases of limited commitment: one in which the Principal commits exclusively to actions (mandated standards), and another in which she commits exclusively to inspection. This analysis is driven by two motivations.

²⁰The optimal solution fails to exist when utilities are linear in actions, for example when a acts like a transfer. See [Ahmadzadeh and Waizmann \(2024\)](#), who show that the optimal solution does not exist in such settings.

²¹To resolve the discrepancy between theory and practice regarding extreme rewards, one may either impose an exogenous upper bound on rewards, as in [Border and Sobel \(1987\)](#), or restrict the Principal’s ability to commit to ex-post inefficient actions, as suggested by [Halac and Yared \(2020\)](#). We follow the latter approach in Section 4.2.

Theoretically, disaggregating the Principal’s commitment power allows us to isolate the strategic value of each regulatory instrument. By quantifying the losses from an inability to commit to actions or to inspections, we can assess which dimension of commitment is more important for balancing inspection costs and information rents against surplus extraction.

In practice, regulatory bodies face institutional and political constraints that naturally restrict their commitment capabilities. For instance, an environmental agency might reliably enforce emission standards (commitment to actions) but may be unable to commit to an inspection policy when its inspection effort is unobservable, especially under stochastic inspection, because deviations are difficult to monitor and verify (no commitment to inspections). Conversely, a regulator might possess the capacity to verify firms’ true compliance costs (commitment to inspections) but face political pressure that prevents ex-post enforcement (no commitment to actions).

By formally evaluating these variations in the subsequent subsections, we gain deeper insights into how a Principal can optimally navigate institutional rigidities and political constraints to sustain effective environmental regulation.

4.1 Commitment Only to Actions

From an ex post perspective, the Principal may lack the incentive to conduct inspections. When the Agent sends a message to the Principal, it might not be optimal for the Principal to inspect the Agent, making it challenging to ensure adherence to the contract. Additionally, if the Principal’s inspection efforts are unobservable, she might reduce her effort to save on inspection costs. Especially when the optimal inspection policy is stochastic, it becomes difficult to monitor whether the Principal is adhering to the contract, and it is hard to enforce and verify deviations in court.²²

²²A prominent example is the United States Environmental Protection Agency (EPA) in its enforcement of the Clean Air Act. The EPA commits to legally binding emission standards codified in federal law. However, it cannot commit to a specific, continuous auditing frequency to verify privately known abatement costs. This limitation stems fundamentally from budgetary volatility—enforcement funds are dictated by highly politicized annual congressional appropriations—as well as shifting executive priorities across different presidential administrations. Because the agency cannot guarantee future funding levels or the continuous deployment of specialized inspectors, it cannot credibly promise to audit a fixed percentage of firms, often forcing a reliance on alternative frameworks such as its “Audit Policy” for self-policing. See the United States Environmental Protection Agency’s “Incentives for Self-Policing: Discovery, Disclosure,

In this section, we examine a scenario where the Principal can commit to the mandated actions—both the uninspected action a and the post-inspection action a^I —but cannot commit to an inspection policy i .

Formally, the Principal publicly commits to the action mappings $a(\cdot) : M \rightarrow \mathbb{R}_+$ and $a^I(\cdot, \cdot) : M \times [\underline{\theta}, \bar{\theta}] \rightarrow \mathbb{R}_+$, which are the fixed instruments. The induced game is as follows. The Agent privately observes his type $\theta \sim F(\cdot)$ and sends a message m . Upon receiving m , the Principal chooses an inspection probability $i(m) \in [0, 1]$. If inspection occurs, the precommitted action $a^I(m, \theta)$ is mandated; otherwise, the precommitted action $a(m)$ is mandated. The Agent then accepts or rejects the mandated action. The Agent’s strategy is $\sigma(\cdot \mid \theta) \in \Delta(M)$. The Principal’s strategy is $i(m) \in [0, 1]$, and her belief is $\beta(\theta \mid m)$. Strategies and beliefs are measurable.

Proposition 2 *The Principal’s preferred equilibrium when she commits only to actions is outcome equivalent to the optimal full-commitment mechanism restricted to deterministic inspection ($i(\theta) \in \{0, 1\}$).*

We first characterize the optimal full-commitment mechanism when the Principal is restricted to deterministic inspection. This benchmark policy partitions the type space using two thresholds, θ_c^D and θ_e^D .

Because randomized inspection is ruled out by assumption, the stochastic inspection zone is replaced by a full inspection region. For the most efficient types ($\theta < \theta_c^D$), the Principal conducts no inspections and mandates a constant uninspected action capped at $a(\theta) = \theta_c^D$, preserving a structure similar to that of the unrestricted full-commitment case. However, for intermediate types ($\theta \in [\theta_c^D, \theta_e^D]$), the policy transitions to full inspection ($i = 1$) where the Principal mandates her preferred action, $a^I(\theta, \theta) = \theta$. Unlike under unrestricted full commitment, where the no-inspection action satisfies $a(\theta) = a_c \geq \theta_c$, full inspection allows the Principal to set the action for efficient types at the cutoff, $a(\theta) = \theta_c^D$. Consequently, all types above θ_e^D receive zero payoff. Finally, for inefficient types ($\theta > \theta_e^D$), the Principal

Correction and Prevention of Violations” (commonly known as the EPA Audit Policy).

excludes them by mandating the action $\underline{\theta}$.²³ Figure 4 illustrates this mechanism.²⁴

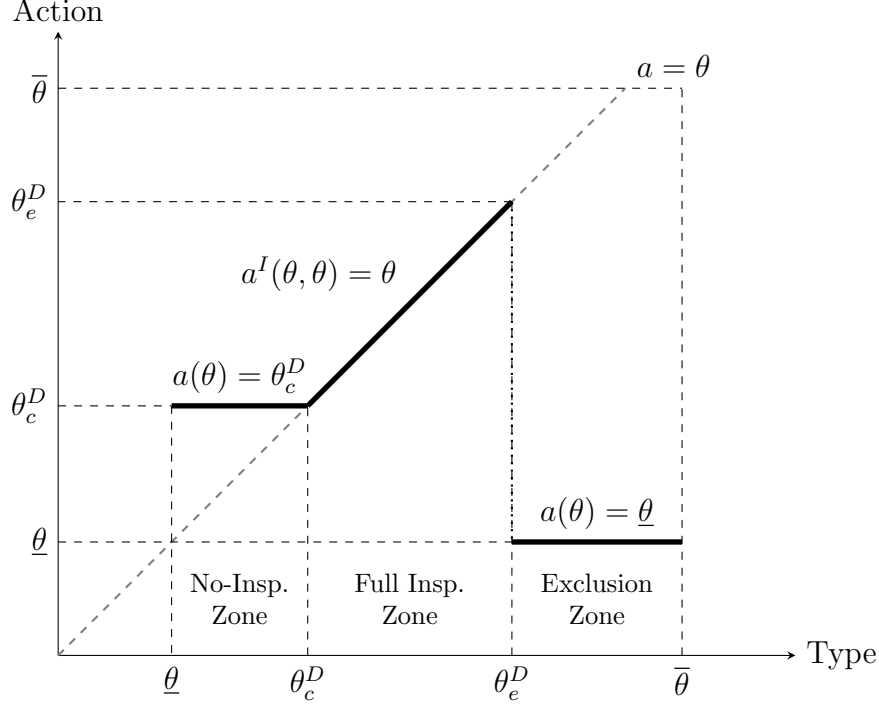


Figure 4: Optimal full-commitment mechanism restricted to deterministic inspection.

²³The threshold structure of the optimal policy aligns with deterministic findings in [Townsend \(1979\)](#), [Gale and Hellwig \(1985\)](#), [Malenko \(2019\)](#), and [Halac and Yared \(2020\)](#). However, our environment generates distinct predictions. First, unlike the aforementioned models, our framework generates an exclusion region because the first-best value of high types is decreasing, and the Principal directly bears the inspection cost. Second, because our Agent is protected by an ex-post participation constraint rather than an ex-ante individual rationality constraint, inspection does not serve as insurance against excessive losses.

²⁴To characterize θ_c^D and θ_e^D , let the problem $\mathbb{P}_D$ be defined as:

$$\mathbb{P}_D : \max_{\theta_c^D \in [\underline{\theta}, \bar{\theta}]} \left\{ \int_{\underline{\theta}}^{\theta_c^D} (v(\theta, \theta_c^D)) dF(\theta) + \int_{\theta_c^D}^{\bar{\theta}} (v(\theta, \theta) - \phi) \mathbb{1}_{v(\theta, \theta) \geq \phi} dF(\theta) \right\}.$$

Let Θ_c^D be the set of solutions to $\mathbb{P}_D$. Problem $\mathbb{P}_D$ is continuous in $\theta_c \in [\underline{\theta}, \bar{\theta}]$; therefore, it admits a maximizer. For any $\theta_c^D \in \Theta_c^D$, define the upper threshold $\theta_e^D \geq \theta_c^D$ such that:

$$\theta_e^D = \begin{cases} \theta_c^D & \text{if } v(\theta_c^D, \theta_c^D) \leq \phi \\ \bar{\theta} & \text{if } v(\bar{\theta}, \bar{\theta}) > \phi \end{cases}$$

Otherwise, define θ_e^D implicitly as the solution to $v(\theta_e^D, \theta_e^D) = \phi$.

We now construct the Principal-preferred equilibrium that achieves this deterministic full-commitment benchmark payoff, relying exclusively on her ability to commit to the actions.

The equilibrium reporting strategy is binary. Types in the no-inspection and exclusion regions, $\theta \in [\underline{\theta}, \theta_c^D) \cup (\theta_e^D, \bar{\theta}]$, send message m_N , while intermediate types in the full-inspection region, $\theta \in [\theta_c^D, \theta_e^D]$, send message m_I . The Principal commits to an uninspected action cap of $a(m_N) = \theta_c^D$ for message m_N , and a maximally punitive action $a(m) = \underline{\theta}$ for all $m \neq m_N$. Upon inspecting any type, she mandates her preferred action $a^I(m, \theta) = \theta$.²⁵ The Principal's equilibrium inspection strategy is to inspect after receiving m_I ($i(m_I) = 1$) and to never inspect otherwise ($i(m) = 0$ for all $m \neq m_I$). To deter off-path deviations, if the Principal receives any unexpected message $m \notin \{m_N, m_I\}$, her belief $\beta(\theta|m)$ assigns probability one to the lowest type $\underline{\theta}$.

This strategy profile constitutes a Perfect Bayesian Equilibrium. First, consider the Agent's incentives. Efficient types ($\theta < \theta_c^D$) have no incentive to deviate to m_I ; doing so triggers an inspection yielding the efficient action $a^I = \theta$, netting them a payoff of zero, which is strictly worse than the positive informational rent they earn by accepting their designated cap θ_c^D . Conversely, intermediate and inefficient types ($\theta \geq \theta_c^D$) receive a payoff of zero under the equilibrium strategy—either through full inspection (m_I) or by rejecting the uninspected cap θ_c^D (m_N). Because no deviation yields a strictly positive payoff, they are willing to comply with the proposed strategy.

The Principal remains sequentially rational in her inspection decisions despite lacking ex-ante commitment to an audit policy. To force herself to inspect the intermediate types reporting m_I , she utilizes strategic self-punishment by committing to the “toxic” uninspected action $a(m_I) = \underline{\theta}$. If she deviates and forgoes the inspection to save the fixed cost ϕ , she would be forced to mandate $\underline{\theta}$, which the Agent would reject, yielding her a payoff of zero. Because the net surplus of inspecting intermediate types is strictly positive ($v(\theta, \theta) - \phi \geq 0$), she strictly prefers to conduct the audit. Conversely, upon receiving m_N , zero inspection ($i(m_N) = 0$) is sequentially rational. Suppose, for the sake of contradiction, that the Principal strictly prefers to inspect after receiving m_N . Then an alternative mechanism that mandates universal inspection for all types (which is trivially incentive-compatible) would yield a strictly higher ex-ante expected payoff than the

²⁵Because mandating the efficient action is sequentially rational for the Principal, commitment power over $a^I(\cdot, \cdot)$ is not necessary to sustain this equilibrium.

proposed equilibrium. However, the proposed equilibrium perfectly replicates the payoff of the optimal deterministic full-commitment mechanism. By definition, no alternative deterministic mechanism—including universal inspection—can strictly dominate it. Therefore, the expected net value of inspecting the pooled efficient and inefficient types in m_N must be weakly negative, rendering the decision not to inspect sequentially rational.

Although equilibria with stochastic inspection ($i \in (0, 1)$) for intermediate types exists, they cannot give the Principal a strictly higher payoff than the equilibrium described above. In any equilibrium, types segregate by efficiency due to single-crossing properties. If a specific type is not inspected on the equilibrium path, all more efficient types must also remain uninspected. Similarly, if a type is excluded, all less efficient types must also be excluded. This property partitions the type space into three ordered regimes: a no-inspection zone for efficient types, a positive-inspection zone for intermediate types, and an exclusion zone for inefficient types. To randomize ex-post, the Principal must be indifferent between inspecting and not inspecting, meaning her expected payoff from randomizing is equivalent to her payoff from inspecting with probability one. By inspecting intermediate types with probability one, the Principal minimizes information rents, allowing her to tighten the uninspected action for efficient types exactly to the boundary threshold θ_c^D . Consequently, the Principal can always “purify” any stochastic equilibrium into our deterministic binary structure without losing payoff. Thus, this equilibrium exhausts the Principal’s rent extraction capabilities under limited commitment, rendering it her preferred PBE.

Our framework provides a distinct modeling perspective compared to the Costly State Verification literature without commitment, particularly [Khalil \(1997\)](#). In Khalil’s model, the Agent undertakes a sunk, observable productive action before the Principal makes her audit decision, yielding a game of costly signaling. Within this timing structure, the Principal must rely on stochastic inspection in equilibrium to deter noncompliance. In stark contrast, our model operates strictly as a game of cheap talk: the Agent sends a costless report, the Principal chooses her inspection strategy, and only then are the state-contingent actions mandated. By retaining ex ante commitment to these actions, the principal constructs a prohibitive uninspected action that serves as a mechanism of strategic self-punishment, perfectly aligning her ex-post incentives to audit. Consequently, whereas the signaling environment predicts an equilibrium characterized by stochastic auditing, the Principal-preferred equilibrium in our cheap-talk framework relies entirely on deterministic inspection.

Melumad and Mookherjee (1989) also consider a setting with no commitment to inspection, but where the Principal maintains commitment to transfers. A crucial difference in our environment is that our Agent is protected by ex-post participation constraints. In our model, the Principal strategically utilizes this constraint to her advantage. By committing to an uninspected action $a(m_I) = \underline{\theta}$ that she knows the Agent will certainly reject, she artificially lowers the ex-post value of not inspecting, thereby making her threat to inspect credible without generating a massive equilibrium loss for the Agent. Therefore, in our model, the Principal can achieve the full deterministic inspection payoff despite her lack of commitment to the audit itself.

4.2 Commitment Only to Inspection

In this section, we examine a scenario where a Principal commits to an inspection policy but does not commit to actions. For instance, consider an inspection agency that outlines the conditions under which inspections occur and provides this information to a regulatory body. However, the regulatory body does not set specific standards.²⁶

Suppose the Principal only commits to $i(\cdot) : \mathbb{M} \rightarrow [0, 1]$, which is the fixed instrument. The induced game is as follows. The Agent privately observes his type $\theta \sim F(\cdot)$, and sends a message m . Inspection occurs or not based on precommitted $i(m)$. With inspection, she observes the type of the Agent and mandates an action a^I . Without inspection, she mandates an action a . The Agent accepts or rejects the mandated standard. The Agent's strategy is $m(\theta) \in \mathbb{R}_+$. The Principal's strategy is $(a(m) \in \mathbb{R}_+, a^I(m, \theta) \in \mathbb{R}_+)$. The Principal's belief is $\beta(\theta|m)$. Strategies

²⁶The European Union Emissions Trading System (EU ETS) during its Phase I pilot period (2005–2007) exemplifies this dynamic. The EU enforced absolute commitment to verification: participating installations were subject to stringent, legally mandated Monitoring, Reporting, and Verification (MRV) protocols conducted by accredited third-party auditors. However, the system suffered from a severe lack of commitment to the actual regulatory standard (the total emission cap). Once initial audits revealed the true baseline emissions and the massive potential abatement costs to heavy industry, national governments faced intense political pressure. Fearing job losses, industrial bankruptcy, and carbon leakage to jurisdictions outside the EU, regulators caved to industry lobbying. Because they could not politically commit to imposing crippling costs ex-post, they relaxed the standard by massively over-allocating free emission allowances, ultimately causing the price of carbon to crash to near zero. For a comprehensive review of the EU ETS Phase I, the over-allocation of allowances, and the underlying political economy constraints, see Ellerman, A. D., and Joskow, P. L. (2008), *The European Union's Emissions Trading System in Perspective*, Pew Center on Global Climate Change.

and beliefs are measurable. An immediate observation is that the Principal is opportunistic in the case of inspection: $a^I(m, \theta) = \theta$.

Lemma 1 *Fix $i(\cdot)$. In any PBE of the game in which the Principal commits only to inspection, (i) $a(m)$ is constant for all on-path messages m such that $i(m) < 1$. (ii) For on-path messages m_1 and m_2 , if $i(m_1) < 1$ and $i(m_2) < 1$, then $i(m_1) = i(m_2)$.*

Lemma 1 has two implications. First, all on-path messages that are inspected with probability less than one must induce the same action when inspection does not occur. The key reason is that, after a message m , the Principal chooses the no-inspection action optimally given her posterior belief. For every on-path message with $i(m) < 1$, there are types sending that message who are arbitrarily close to $a(m)$ from below. If not the Principal could lower $a(m)$ slightly, and increase her payoff. Now suppose two such messages, m_1 and m_2 , induced different no-inspection actions, say $a(m_1) > a(m_2)$. Types who send m_2 and are very close to $a(m_2)$ from below obtain almost zero payoff. But because $i(m_1) < 1$, they can mimic m_1 and, with positive probability, receive the strictly larger action $a(m_1)$. This gives them a strictly positive deviation payoff, contradicting equilibrium. Therefore $a(m_1) = a(m_2)$ for all on-path messages with inspection probability less than one.

Second, once these no-inspection actions are the same, the inspection probabilities must also be the same. If two on-path messages m_1 and m_2 have the same no-inspection action but $i(m_1) > i(m_2)$, then a type sending m_1 who accepts the common no-inspection action strictly prefers to send m_2 . The action obtained when no inspection occurs is the same, but the probability of receiving it is higher under m_2 . When inspection occurs, the Principal observes the type and chooses the efficient action, which gives the Agent zero rent. Hence a lower inspection probability is strictly better for any type who benefits from the common no-inspection action. This contradicts equilibrium, so all on-path messages with inspection probability less than one must have the same inspection probability.

As a result of Lemma 1, all on-path messages with inspection probability less than one can be consolidated into a single message, denoted by m_N , without changing equilibrium payoffs. Similarly, all on-path messages inspected with probability one can be consolidated into a single message, denoted by m_I . Therefore, it is without loss to focus on equilibria with at most two messages on the equilibrium path. The following corollary states this formally.

Corollary 1 *For any equilibrium, there exists an outcome-equivalent equilibrium with at most two different messages, m_I and m_N . $i(m_I) = 1$, $i(m_N) < 1$.*

An interpretation of these two messages in Corollary 1 is that when the Agent sends m_I , he is requesting an inspection, whereas when he sends m_N , he is not requesting an inspection but might still be inspected.

Proposition 3 *Suppose the Principal commits only to the inspection policy. Then the Principal's preferred equilibrium contains two thresholds $\underline{\theta} \leq \tilde{\theta}_c^D \leq \tilde{\theta}_e^D \leq \bar{\theta}$ such that types in $[\underline{\theta}, \tilde{\theta}_c^D] \cup [\tilde{\theta}_e^D, \bar{\theta}]$ send m_N , types in $[\tilde{\theta}_c^D, \tilde{\theta}_e^D]$ send m_I , $i(m_N) = 0$, $i(m_I) = 1$, and $a(m_N) = \tilde{\theta}_c^D$. After inspection, the Principal mandates the ex-post optimal action.*

Moreover, relative to the optimal deterministic full-commitment benchmark, the Principal excludes a weakly smaller set of types: $\theta_e^D \leq \tilde{\theta}_e^D$. If this benchmark has no exclusion region, equivalently $\phi < v(\bar{\theta}, \bar{\theta})$, then the Principal excludes no types in her preferred equilibrium, and the equilibrium is outcome-equivalent to the benchmark.

Proposition 3 shows that, when the Principal commits only to inspection, the preferred PBE has a threshold structure similar to that of the optimal deterministic full-commitment mechanism, although the thresholds may differ.

It suffices to consider deterministic inspection. By Lemma 1, it is enough to consider equilibria with two on-path messages, m_N and m_I , where m_I is the inspected message. Once the set of types sending m_N and the action $a(m_N)$ are fixed, the incentive constraints and the optimality of $a(m_N)$ do not depend on the exact value of $i(m_N)$. The inspection probability $i(m_N)$ affects the Principal's payoff linearly:

$$\int_{\theta \in \Theta: m(\theta) = m_N} \left[(1 - i(m_N))v(\theta, a(m_N))\mathbb{1}_{\{a(m_N) \geq \theta\}} + i(m_N)(v(\theta, \theta) - \phi) \right] dF(\theta).$$

Moreover, changing $i(m_N)$ does not trigger deviations by the Agent. Therefore, an interior value $0 < i(m_N) < 1$ cannot be uniquely optimal. Hence, the preferred PBE involves deterministic inspection.

The interval structure follows from a sorting argument. Once the no-inspection cap $a(m_N) = \tilde{\theta}_c^D$ is fixed, any inspected type must lie above this cap; otherwise he would prefer to send m_N and receive a positive rent. Now suppose a higher type is

inspected while a lower type above $\tilde{\theta}_c^D$ remains in the no-inspection message. Since this lower type is above the cap, he rejects the no-inspection action and is therefore excluded. Switching these two types weakly raises the payoff from inspection, because $v(\theta, \theta)$ is decreasing in θ . This switch also preserves the optimality of the cap after m_N . Fix any alternative action $\tilde{s}$. If $\tilde{s} \leq \tilde{\theta}_c^D$, the switched types reject both $\tilde{s}$ and the cap, so their contribution to the payoff comparison is unchanged. If $\tilde{s} > \tilde{\theta}_c^D$, then before the switch the Principal had already chosen not to replace the cap by $\tilde{s}$. After the switch, $\tilde{s}$ becomes weakly less attractive: the no-inspection posterior replaces a lower type with a higher type, who either rejects $\tilde{s}$ or, if he accepts it, is less valuable to serve. Hence no alternative action becomes more profitable than the cap, so $a(m_N) = \tilde{\theta}_c^D$ remains optimal after m_N .²⁷

The comparison with the deterministic inspection benchmark is driven by the upper threshold. If $\tilde{\theta}_e^D < \theta_e^D$, then the types in $[\tilde{\theta}_e^D, \theta_e^D]$ are assigned to the no-inspection message, even though the deterministic benchmark would inspect them. Moving these types to the inspection region weakly increases the Principal's payoff. Furthermore, after this change, the Principal's best reply $a(m_N)$ remains unchanged, as the posterior distribution over m_N is unchanged up to a common rescaling.

Finally, if $\phi < v(\bar{\theta}, \bar{\theta})$, then the optimal deterministic inspection policy has no exclusion region, so $\theta_e^D = \bar{\theta}$. Since $\theta_e^D \leq \tilde{\theta}_e^D$, it follows that $\tilde{\theta}_e^D = \bar{\theta}$. In this case, the equilibrium under commitment only to inspection is outcome-equivalent to the optimal deterministic inspection policy.

This environment is related to [Halac and Yared \(2020\)](#), who study a delegation model in which the Principal can commit to inspection but may revise the action. Their analysis focuses on deterministic inspections, and their payoff environment has $v(\theta, \theta)$ increasing in θ . As a result, their model does not feature the upper-tail exclusion region that appears here.

4.3 Comparing Different Commitment

This section summarizes how the Principal's payoff and policy structure vary across commitment regimes. Full commitment weakly dominates the limited-commitment regimes because the Principal can commit both to the inspection rule and to the actions imposed after each message. The gain is strict whenever the

²⁷Under this switch, the Principal's payoff is unchanged because both switched types reject the cap.

deterministic full-commitment benchmark uses inspection for a positive mass of types. In that case, the unrestricted mechanism cannot avoid stochastic inspection by using full inspection, since full-inspection regions are ruled out by Proposition 1. Instead, the Principal uses stochastic inspection together with post-inspection rewards to reduce information rents. Hence full commitment yields a strictly higher payoff than the deterministic-inspection benchmark, which is also the Principal's preferred PBE when she commits only to actions.

When the Principal commits only to inspection, she faces an additional sequential constraint: after a no-inspection message, the action must be optimal given her posterior belief. This constraint may force her to inspect some inefficient types that would otherwise be excluded, in order to make exclusion of even less efficient types optimal without inspection. As a result, her payoff is weakly below the deterministic-inspection benchmark, and therefore weakly below the payoff obtained when she commits only to actions. If the deterministic benchmark has no exclusion region, this additional constraint is not binding, and the two limited-commitment regimes are outcome-equivalent.

5 Extensions

5.1 No Commitment

Now we analyze the case in which the Principal has no commitment power. The timing is as follows. The Agent privately observes his type $\theta \sim F(\cdot)$ and sends a message m . After observing m , the Principal decides whether to inspect. If she inspects, she observes the true type and chooses the post-inspection action. If she does not inspect, she chooses the no-inspection action. Finally, the Agent accepts or rejects the mandated standard.

The Agent's strategy is $m(\theta) \in \mathbb{R}_+$. The Principal's strategy is

$$(i(m) \in [0, 1], a(m) \in \mathbb{R}_+, a^I(m, \theta) \in \mathbb{R}_+).$$

The Principal's belief is $\beta(\theta|m)$. Since the Principal has no commitment power, after inspection she chooses the sequentially rational action, $a^I(m, \theta) = \theta$.

By Lemma 1, all on-path messages with inspection probability less than one can be consolidated into one no-inspection message, denoted m_N . We focus on finite-interval strategies, where the Agent's message rule alternates between m_N

and the inspection message m_I on finitely many intervals.

Proposition 4 *Suppose the Principal does not have commitment power. Within the class of finite-interval equilibria, the Principal's preferred equilibrium has two thresholds $\underline{\theta} \leq \theta_c^{NC} \leq \theta_e^{NC} \leq \bar{\theta}$. Types in $[\underline{\theta}, \theta_c^{NC}] \cup [\theta_e^{NC}, \bar{\theta}]$ send m_N , while types in $[\theta_c^{NC}, \theta_e^{NC}]$ send m_I . After m_N , the Principal does not inspect and mandates $a(m_N) = \theta_c^{NC}$. After m_I , the Principal inspects and mandates $a^I(m_I, \theta) = \theta$.*

[Banks \(1989\)](#) examines a model in which an Agent requests a budget. In their open-procedure environment, the Principal may inspect or not and chooses the Agent's budget accordingly. Utilities are linear, and inspection is imperfect: with positive probability, inspection does not reveal the Agent's information. Because inspection is imperfect, types just below the threshold analogous to θ_c^{ND} find it profitable to send the message intended for inspected types, m_I . As a result, full pooling is the only equilibrium in their setting, and no information is transmitted from the Agent to the Principal.

In our environment, inspection perfectly reveals the Agent's type. Therefore, if a type below θ_c^D sends m_I , inspection identifies the misreport and the Principal can impose the punishment action $\underline{\theta}$. This prevents low types from mimicking the inspected pool. With perfect inspection, the semi-separating structure generated by m_N and m_I is therefore sustainable.

5.2 Partial Commitment

We now consider an intermediate commitment environment. The Principal cannot commit to an arbitrary inspection rule, but she can commit to inspect whenever the Agent requests an inspection. This is a natural restriction in applications where institutional rules prevent a regulator from refusing an audit once the regulated party has requested one. Thus, the Agent can always choose a message that triggers inspection. We show that this limited commitment is enough to implement the optimal deterministic full-commitment payoff, provided inspection is not too costly.

Proposition 5 *Suppose $\phi < v(\bar{\theta}, \bar{\theta})$, and suppose the Principal commits to inspect after message m_I . Then the Principal's preferred equilibrium is outcome-equivalent to the optimal deterministic full-commitment mechanism.*

The intuition is as follows. Under partial commitment, the Principal cannot commit to an arbitrary inspection rule, but she can commit to inspect after the

audit-request message m_I . Since $\phi < v(\bar{\theta}, \bar{\theta})$ and $v(\theta, \theta)$ is decreasing in θ , in the optimal deterministic full-commitment mechanism, the exclusion threshold is empty. The deterministic mechanism has two messages. Types $\theta \leq \theta_c^D$ send the no-inspection message m_N , while types $\theta > \theta_c^D$ send the inspection message m_I .

The Principal is committed to inspect after m_I , so the only sequential incentive to check is what she does after m_N . After observing m_N , her posterior is supported on $[\underline{\theta}, \theta_c^D]$. Hence the best uninspected action does not exceed θ_c^D : if it does, lowering it to θ_c^D would keep all types in the posterior accepting the action and would raise the Principal's payoff. It also cannot be strictly below θ_c^D . If some $a(m_N) < \theta_c^D$ gave a higher payoff, then a deterministic full-commitment policy with cap $a(m_N)$ for the low types and inspection for all higher types would generate a higher value than the optimal deterministic mechanism, which contradicts the definition of θ_c^D . Thus the Principal optimally chooses $a(m_N) = \theta_c^D$.

Finally, the Principal has no incentive to inspect after m_N . If inspecting the types who send m_N were better than mandating them the cap θ_c^D , then inspecting all types would again yield a higher payoff than the optimal deterministic full-commitment payoff. Thus no inspection after m_N is sequentially optimal. The Agent's incentives are also satisfied: low types prefer the cap θ_c^D to requesting inspection and receiving the Principal's preferred action, while high types cannot profit from sending m_N because $\theta_c^D < \theta$ and they reject that action. Hence the two-message equilibrium implements the payoff of the optimal deterministic full-commitment mechanism.

6 Conclusion

This paper studies how costly inspection and ex post participation shape regulation when transfers are unavailable. Under full commitment, the optimal mechanism has up to three ordered regions: efficient types are pooled without inspection, intermediate types are inspected stochastically, and high types are excluded. In the stochastic-inspection region, inspection serves not only to detect mimicking but also to reward truthful reporting. The action assigned after truthful verification is higher than the uninspected action, which is itself above the Principal's ex post preferred action. Nevertheless, full inspection is never optimal on a set of positive measure because its informational benefit can be preserved while reducing the fixed inspection cost. The optimal policy therefore balances inspection costs, information rents, action distortions, and the surplus forgone

through exclusion.

The comparison across commitment environments reveals a sharp asymmetry. Commitment only to actions implements the optimal deterministic full-commitment outcome: by committing to a self punishing fallback action, the Principal can make inspection sequentially optimal without committing directly to inspect. Commitment only to inspection is generally weaker because the uninspected action must be optimal given the Principal's posterior, which introduces an additional sequential-rationality constraint. Consequently, the Principal's preferred equilibrium under commitment only to actions equals the deterministic full-commitment value and weakly exceeds her value under commitment only to inspection. When exclusion is absent, this difference disappears. Moreover, if inspection cost is not high, the limited ability to commit to inspection upon the Agent's request is sufficient to implement the optimal deterministic outcome.

Appendix

Assumption 1 Generalized Utilities

- (i) **Smoothness.** The functions $u(\theta, a)$ and $v(\theta, a)$ are continuous in (θ, a) on the relevant domain. They are twice continuously differentiable in a on the region $a > \theta$.
- (ii) **Agent preferences and ex-post participation.** The Agent's utility is strictly increasing and strictly concave in the action:

$$u_a(\theta, a) > 0, \quad u_{aa}(\theta, a) < 0.$$

The outside option is normalized by

$$u(\theta, \theta) = 0.$$

Moreover,

$$u(\theta, a) > 0 \quad \text{if } a > \theta, \quad u(\theta, a) < 0 \quad \text{if } a < \theta,$$

and

$$\lim_{a \rightarrow \infty} u(\theta, a) = \infty$$

uniformly in θ .

- (iii) **Log-supermodularity.** The Agent's utility is log-supermodular: for every $\theta' < \theta$ and every $a' > a > \theta$,

$$\frac{u(\theta', a')}{u(\theta', a)} \leq \frac{u(\theta, a')}{u(\theta, a)}.$$

- (iv) **Principal preferences.** The Principal's utility is strictly decreasing and strictly concave in the action on the region $a > \theta$:

$$v_a(\theta, a) < 0, \quad v_{aa}(\theta, a) < 0.$$

The first-best valuation is positive and strictly decreasing in θ :

$$v(\theta, \theta) > 0, \quad \frac{d}{d\theta} v(\theta, \theta) < 0.$$

Finally,

$$\lim_{a \rightarrow \infty} v(\theta, a) = -\infty$$

uniformly in θ .

(v) **Boundary flatness.** At the efficient action,

$$v_a(\theta, \theta) = 0 \quad \forall \theta \in [\underline{\theta}, \bar{\theta}],$$

and there exist constants $K < \infty$, $\gamma > 1$, and $\eta > 0$ such that

$$0 \leq v(\theta, \theta) - v(\theta, \theta + d) \leq Kd^\gamma \quad \forall d \in [0, \eta], \quad \forall \theta \in [\underline{\theta}, \bar{\theta}].$$

(vi) **Cost of utility delivery.** The Principal's marginal cost of delivering utility to the Agent,

$$MC(\theta, a) \equiv -\frac{v_a(\theta, a)}{u_a(\theta, a)},$$

is strictly increasing in a on the region $a > \theta$ and satisfies

$$\lim_{a \rightarrow \infty} MC(\theta, a) = \infty$$

uniformly in θ .

Lemma 2 *The payoff functions*

$$u(\theta, a) = (a - \theta + 1)^\alpha - 1, \quad v(\theta, a) = v_0(\theta) - (a - \theta)^\beta,$$

with $0 < \alpha < 1 < \beta$, satisfy Assumption [1i-1vi](#), provided that $v_0(\theta) > 0$ is continuous and strictly decreasing in θ .

Proof. First,

$$u_a(\theta, a) = \alpha(a - \theta + 1)^{\alpha-1} > 0,$$

and

$$u_{aa}(\theta, a) = \alpha(\alpha - 1)(a - \theta + 1)^{\alpha-2} < 0,$$

because $0 < \alpha < 1$. Moreover,

$$u(\theta, \theta) = 1^\alpha - 1 = 0.$$

If $a > \theta$, then $a - \theta + 1 > 1$, so $u(\theta, a) > 0$. If $a < \theta$ and $a - \theta + 1 > 0$, then

$a - \theta + 1 < 1$, so $u(\theta, a) < 0$. Finally,

$$\lim_{a \rightarrow \infty} u(\theta, a) = \infty.$$

Thus Assumption [lii](#) is satisfied.

Next, for fixed $a' > a > \theta$,

$$\frac{\partial}{\partial \theta} \log \left(\frac{u(\theta, a')}{u(\theta, a)} \right) = -\frac{\alpha(a' - \theta + 1)^{\alpha-1}}{(a' - \theta + 1)^\alpha - 1} + \frac{\alpha(a - \theta + 1)^{\alpha-1}}{(a - \theta + 1)^\alpha - 1}.$$

The function

$$z \mapsto \frac{\alpha z^{\alpha-1}}{z^\alpha - 1}$$

is strictly decreasing for $z > 1$, since

$$\frac{d}{dz} \left(\frac{\alpha z^{\alpha-1}}{z^\alpha - 1} \right) = -\frac{\alpha z^{\alpha-2}(z^\alpha + \alpha - 1)}{(z^\alpha - 1)^2} < 0.$$

Because

$$a' - \theta + 1 > a - \theta + 1 > 1,$$

we obtain

$$\frac{\partial}{\partial \theta} \log \left(\frac{u(\theta, a')}{u(\theta, a)} \right) > 0.$$

Hence, for every $\theta' < \theta$ and $a' > a > \theta$,

$$\frac{u(\theta', a')}{u(\theta', a)} \leq \frac{u(\theta, a')}{u(\theta, a)}.$$

Therefore Assumption [liii](#) is satisfied.

For the Principal,

$$v_a(\theta, a) = -\beta(a - \theta)^{\beta-1} < 0 \quad \text{for } a > \theta,$$

and

$$v_{aa}(\theta, a) = -\beta(\beta - 1)(a - \theta)^{\beta-2} < 0 \quad \text{for } a > \theta.$$

Also,

$$v(\theta, \theta) = v_0(\theta) > 0,$$

which is strictly decreasing by assumption. Thus Assumption [liv](#) is satisfied.

Moreover,

$$v_a(\theta, \theta) = 0,$$

and

$$v(\theta, \theta) - v(\theta, \theta + d) = d^\beta.$$

Hence Assumption 1v holds with $\gamma = \beta > 1$.

Finally,

$$MC(\theta, a) = -\frac{v_a(\theta, a)}{u_a(\theta, a)} = \frac{\beta}{\alpha}(a - \theta)^{\beta-1}(a - \theta + 1)^{1-\alpha}.$$

For $a > \theta$,

$$\frac{\partial}{\partial a} MC(\theta, a) = MC(\theta, a) \left[\frac{\beta - 1}{a - \theta} + \frac{1 - \alpha}{a - \theta + 1} \right] > 0,$$

because $0 < \alpha < 1 < \beta$. Also,

$$\lim_{a \rightarrow \infty} MC(\theta, a) = \infty.$$

Therefore Assumption 1vi is satisfied. ■

Proposition 6 *Suppose Assumption 1i, 1ii, 1iv, and 1vi hold. Then there exists a finite upper bound*

$$\bar{A} < \infty$$

such that it is without loss of generality to restrict all assigned actions to

$$[\underline{\theta}, \bar{A}].$$

Consequently, an optimal full-commitment mechanism exists.

Proof. First, consider the feasible mechanism in which the Principal inspects every type with probability one and assigns the efficient action:

$$i(\theta) = 1, \quad a^I(\theta, \theta) = \theta.$$

By Assumptions 1i, and 1iv, this mechanism gives the Principal a finite expected payoff

$$\underline{V} = \int_{\underline{\theta}}^{\bar{\theta}} [v(\theta, \theta) - \phi] f(\theta) d\theta > -\infty.$$

Therefore, any optimal mechanism must give the Principal expected payoff at least $\underline{V}$.

By Assumption 1ii, for each fixed θ , the map

$$a \mapsto u(\theta, a)$$

is strictly increasing and unbounded above. Hence, for every utility level $r \geq 0$, there is a unique action, denoted $a_\theta(r)$, satisfying

$$u(\theta, a_\theta(r)) = r.$$

Define

$$\Psi_\theta(r) = v(\theta, a_\theta(r)).$$

By Assumptions 1i and 1ii,

$$\Psi'_\theta(r) = \frac{v_a(\theta, a_\theta(r))}{u_a(\theta, a_\theta(r))} = -MC(\theta, a_\theta(r)).$$

By Assumption 1vi, $MC(\theta, a)$ is strictly increasing in a and diverges as $a \rightarrow \infty$. Therefore Ψ_θ is decreasing, strictly concave, and has arbitrarily steep negative slope for large r .

Fix a type θ and let $a^I(\theta) \equiv a^I(\theta, \theta)$. If $0 < i(\theta) < 1$, the standard variational argument implies that, without loss of generality,

$$a^I(\theta) \geq a(\theta).$$

Indeed, if $a(\theta) > a^I(\theta)$, then increasing $a^I(\theta)$ and decreasing $a(\theta)$ while holding the truthful payoff fixed relaxes all IC constraints and raises the Principal's payoff because $MC(\theta, a)$ is strictly increasing by Assumption 1vi.

Let

$$r_0 = u(\theta, a(\theta)), \quad r_1 = u(\theta, a^I(\theta)).$$

If $r_1 > r_0$, then holding the truthful expected utility fixed requires

$$i(\theta) = \frac{\pi(\theta) - r_0}{r_1 - r_0}.$$

The Principal's payoff from this type can be written as

$$W(r_1) = \Psi_\theta(r_0) + \frac{\pi(\theta) - r_0}{r_1 - r_0} [\Psi_\theta(r_1) - \Psi_\theta(r_0) - \phi].$$

Differentiating,

$$W'(r_1) = \frac{\pi(\theta) - r_0}{(r_1 - r_0)^2} \left[\Psi'_\theta(r_1)(r_1 - r_0) - (\Psi_\theta(r_1) - \Psi_\theta(r_0)) + \phi \right].$$

Since $\Psi'_\theta(r) = -MC(\theta, a_\theta(r))$, the bracketed term is

$$- \int_{r_0}^{r_1} \left[MC(\theta, a_\theta(r_1)) - MC(\theta, a_\theta(s)) \right] ds + \phi.$$

By Assumption 1vi, this expression becomes strictly negative for all sufficiently large r_1 , uniformly in θ . Hence very large $a^I(\theta)$ strictly reduce the Principal's payoff. Thus there exists $\bar{A} < \infty$ such that all accepted actions in an optimal mechanism lie in

$$[\underline{\theta}, \bar{A}].$$

Finally, restrict attention to mechanisms satisfying

$$i(\theta) \in [0, 1], \quad a(\theta), a^I(\hat{\theta}, \theta) \in [\underline{\theta}, \bar{A}].$$

the feasible set is compact and the Principal's objective is upper semicontinuous. The feasible set is nonempty because the full-inspection mechanism described above is feasible. Therefore, by Weierstrass' theorem, the Principal's expected payoff attains a maximum. Hence an optimal full-commitment mechanism exists. ■

Proposition 7 *Suppose the Principal possesses full commitment power and Assumption 1 holds. Then the optimal mechanism*

$$\mathbb{M} = (i(\cdot), a(\cdot), a^I(\cdot, \cdot))$$

is characterized, up to null sets and payoff-irrelevant boundary assignments, by two thresholds

$$\underline{\theta} \leq \theta_c \leq \theta_e \leq \bar{\theta}.$$

The mechanism has the following structure:

(i) **No-Inspection Region** ($\theta < \theta_c$): *The Principal does not inspect,*

$$i(\theta) = 0,$$

and mandates a constant uninspected action

$$a(\theta) = a_c \geq \theta_c.$$

(ii) **Stochastic Inspection Region** ($\theta \in [\theta_c, \theta_e]$): The Principal conducts randomized inspections,

$$0 < i(\theta) < 1.$$

In this region,

$$a^I(\theta, \theta) > a(\theta) > \theta.$$

(iii) **Exclusion Region** ($\theta > \theta_e$): The Principal does not inspect,

$$i(\theta) = 0,$$

and mandates a prohibitive action

$$a(\theta) = \underline{\theta},$$

which the Agent rejects.

Proof. By Proposition 6, an optimal full-commitment mechanism exists. Fix an optimal mechanism

$$\mathbb{M} = (i(\cdot), a(\cdot), a^I(\cdot, \cdot)).$$

Under full commitment, after every detected misreport the Principal can impose the maximal punishment. Hence, for every $\hat{\theta} \neq \theta$, she sets

$$a^I(\hat{\theta}, \theta) = \underline{\theta}.$$

By Assumption 1ii, this action is rejected whenever it lies below the Agent's type, so a detected misreport yields payoff zero. Therefore the payoff from mimicking report $\hat{\theta}$ comes only from the no-inspection branch. Incentive compatibility is therefore equivalent to

$$\begin{aligned} & (1 - i(\theta))u(\theta, a(\theta))\mathbb{1}_{\{a(\theta) \geq \theta\}} + i(\theta)u(\theta, a^I(\theta, \theta))\mathbb{1}_{\{a^I(\theta, \theta) \geq \theta\}} \\ & \geq (1 - i(\hat{\theta}))u(\theta, a(\hat{\theta}))\mathbb{1}_{\{a(\hat{\theta}) \geq \theta\}}, \quad \forall \theta, \hat{\theta}. \end{aligned}$$

Moreover, in an optimal mechanism, no type receives more truthful payoff than is needed to satisfy the reduced incentive constraints. Indeed, if the truthful payoff of type θ were strictly above

$$\sup_{\hat{\theta}} (1 - i(\hat{\theta}))u(\theta, a(\hat{\theta}))\mathbb{1}_{\{a(\hat{\theta}) \geq \theta\}},$$

then an accepted truthful action assigned to type θ could be reduced slightly while

preserving all incentive constraints. By Assumption 1ii, this lowers the Agent's payoff continuously, and by Assumption 1iv, it strictly raises the Principal's payoff whenever the reduced action is strictly above θ .

Proof of part (i): No-Inspection Region.

Suppose $\theta_1 < \theta_2$ are two non-excluded types such that

$$i(\theta_1) = i(\theta_2) = 0.$$

Since type θ_2 is non-excluded and not inspected, ex-post participation requires

$$a(\theta_2) \geq \theta_2 > \theta_1.$$

Hence type θ_1 can accept the action assigned to report θ_2 . Incentive compatibility gives

$$u(\theta_1, a(\theta_1)) \geq u(\theta_1, a(\theta_2)).$$

By Assumption 1ii, $u_a > 0$, so

$$a(\theta_1) \geq a(\theta_2).$$

Therefore

$$a(\theta_1) \geq a(\theta_2) \geq \theta_2,$$

so type θ_2 can also accept the action assigned to report θ_1 . Incentive compatibility for type θ_2 implies

$$u(\theta_2, a(\theta_2)) \geq u(\theta_2, a(\theta_1)).$$

Again using Assumption 1ii, we obtain

$$a(\theta_2) \geq a(\theta_1).$$

Thus

$$a(\theta_1) = a(\theta_2).$$

Hence all non-excluded zero-inspection types pool on one common action. Denote this action by a_c .

Next, this zero-inspection pooling set must be a lower interval. Let θ_0 be a non-excluded zero-inspection type with

$$a_c > \theta_0,$$

and fix any $t < \theta_0$. We show that assigning type t the same allocation $i(t) = 0$ and $a(t) = a_c$ preserves incentive compatibility. Take any report $\hat{\theta}$. If $a(\hat{\theta}) < t$, then type t rejects the no-inspection action assigned to report $\hat{\theta}$, so the deviation payoff is zero. If

$$t \leq a(\hat{\theta}) \leq a_c,$$

then Assumption 1ii implies

$$u(t, a_c) \geq u(t, a(\hat{\theta})) \geq (1 - i(\hat{\theta}))u(t, a(\hat{\theta})).$$

Finally suppose

$$a(\hat{\theta}) > a_c.$$

Since $a_c > \theta_0 > t$, Assumption 1iii implies

$$\frac{u(t, a(\hat{\theta}))}{u(t, a_c)} \leq \frac{u(\theta_0, a(\hat{\theta}))}{u(\theta_0, a_c)}.$$

Incentive compatibility of type θ_0 gives

$$u(\theta_0, a_c) \geq (1 - i(\hat{\theta}))u(\theta_0, a(\hat{\theta})),$$

and therefore

$$(1 - i(\hat{\theta}))u(t, a(\hat{\theta})) \leq u(t, a_c).$$

Thus

$$u(t, a_c) \geq (1 - i(\hat{\theta}))u(t, a(\hat{\theta}))\mathbb{1}_{\{a(\hat{\theta}) \geq t\}}, \quad \forall \hat{\theta}.$$

Hence assigning type t the zero-inspection allocation a_c is incentive compatible. It also creates no new deviation opportunity for any other type, because the same no-inspection action a_c was already available through report θ_0 .

It remains to compare the Principal's payoff from type t . Since type t could already mimic report θ_0 , incentive compatibility of the original mechanism implies that the original truthful payoff of type t is at least $u(t, a_c)$. For fixed t , the Principal's marginal cost of delivering utility through an accepted action is

$$MC(t, a) = -\frac{v_a(t, a)}{u_a(t, a)}.$$

By Assumption 1vi, this marginal cost is strictly increasing in a . Since Assumption 1ii gives $u_a > 0$, the Principal's payoff as a function of the delivered utility is strictly concave and decreasing. Hence any lottery over accepted actions that

delivers expected utility at least $u(t, a_c)$ gives the Principal weakly less gross payoff than the deterministic accepted action a_c . Inspection costs make the original allocation even less attractive. Therefore replacing type t 's allocation by $i(t) = 0$ and $a(t) = a_c$ weakly raises the Principal's payoff, and strictly raises it whenever type t was inspected with positive probability. By optimality, every type below a non-excluded zero-inspection type must also belong to the same zero-inspection pool, up to null sets.

Consequently, the non-excluded zero-inspection set is an initial interval. Let its upper endpoint be θ_c . If this set is empty, set $\theta_c = \underline{\theta}$. Since all types in this region accept the common action a_c , ex-post participation and Assumption 1ii imply

$$a_c \geq \theta_c.$$

Thus, up to null sets,

$$i(\theta) = 0, \quad a(\theta) = a_c \geq \theta_c, \quad \theta < \theta_c.$$

Proof of part (ii): Stochastic Inspection Region.

By the no-overdelivery argument above, a type receives strictly positive payoff if and only if there exists a report $\hat{\theta}$ such that

$$1 - i(\hat{\theta}) > 0 \quad \text{and} \quad a(\hat{\theta}) > \theta.$$

If this holds for type θ , then it also holds for every lower type. Therefore the positive-rent set is a lower interval. Let its upper endpoint be θ_e . If this set is empty, set $\theta_e = \theta_c$; if there is no exclusion, set $\theta_e = \bar{\theta}$.

Every type in (θ_c, θ_e) has strictly positive payoff. Such a type cannot have zero inspection. If $i(\theta) = 0$ and $a(\theta) \geq \theta$, then it belongs to the zero-inspection pooling interval already characterized, contradicting $\theta > \theta_c$. If $i(\theta) = 0$ and $a(\theta) < \theta$, then the type rejects the no-inspection action and receives zero payoff, contradicting strict positivity. Hence

$$i(\theta) > 0, \quad \theta \in (\theta_c, \theta_e).$$

We next show that deterministic full inspection cannot occur on a positive measure subset of (θ_c, θ_e) . Suppose, toward a contradiction, that there is a positive-measure set $E \subset (\theta_c, \theta_e)$ such that

$$i(\theta) = 1, \quad \theta \in E.$$

For every $\theta \in E$, strict positivity of the truthful payoff and Assumption 1ii imply

$$a^I(\theta, \theta) > \theta.$$

Moreover, since the truthful payoff equals the largest payoff obtainable from the no-inspection branches, and since no type above θ_e obtains positive payoff, every no-inspection action that is accepted with positive probability is bounded above by θ_e . Hence

$$a^I(\theta, \theta) \leq \theta_e, \quad \theta \in E.$$

By a countable decomposition of E , there exist a number $\rho > 0$ and a positive-measure subset $E_0 \subset E$ such that, for every $\theta \in E_0$,

$$\theta < a^I(\theta, \theta) - \rho \leq \theta_e - \rho$$

and

$$\begin{aligned} & v(\theta, a^I(\theta, \theta) - \rho) - v(\theta, a^I(\theta, \theta)) \\ & - MC(\theta, a^I(\theta, \theta)) \left[u(\theta, a^I(\theta, \theta)) - u(\theta, a^I(\theta, \theta) - \rho) \right] > -\frac{\phi}{2}. \end{aligned}$$

The existence of such a ρ follows from Assumption 1i: as $\rho \downarrow 0$, the left-hand side converges to zero.

For every $\theta \in E_0$, reduce the inspection probability from 1 to $1 - \varepsilon$ and set the no-inspection action equal to

$$a^I(\theta, \theta) - \rho.$$

Let the new inspected action be $a_\varepsilon^I(\theta, \theta)$, chosen to keep the truthful payoff of type θ unchanged:

$$\varepsilon u(\theta, a^I(\theta, \theta) - \rho) + (1 - \varepsilon) u(\theta, a_\varepsilon^I(\theta, \theta)) = u(\theta, a^I(\theta, \theta)).$$

By Assumptions 1i and 1ii, the implicit function theorem gives

$$a_\varepsilon^I(\theta, \theta) = a^I(\theta, \theta) + \varepsilon \frac{u(\theta, a^I(\theta, \theta)) - u(\theta, a^I(\theta, \theta) - \rho)}{u_a(\theta, a^I(\theta, \theta))} + o(\varepsilon).$$

Hence, by a first-order Taylor expansion and $MC(\theta, a) = -v_a(\theta, a)/u_a(\theta, a)$, the

change in the Principal's payoff from type $\theta \in E_0$ is

$$\begin{aligned}\Delta V(\theta) &= \varepsilon v(\theta, a^I(\theta, \theta) - \rho) + (1 - \varepsilon)[v(\theta, a_\varepsilon^I(\theta, \theta)) - \phi] - [v(\theta, a^I(\theta, \theta)) - \phi] \\ &= \varepsilon\phi + \varepsilon \left\{ v(\theta, a^I(\theta, \theta) - \rho) - v(\theta, a^I(\theta, \theta)) \right. \\ &\quad \left. - MC(\theta, a^I(\theta, \theta)) [u(\theta, a^I(\theta, \theta)) - u(\theta, a^I(\theta, \theta) - \rho)] \right\} + o(\varepsilon).\end{aligned}$$

By the choice of E_0 and ρ ,

$$\Delta V(\theta) \geq \frac{\varepsilon\phi}{2} + o(\varepsilon).$$

Therefore, for all sufficiently small ε ,

$$\int_{E_0} \Delta V(\theta) f(\theta) d\theta > 0,$$

because E_0 has positive measure and $f(\theta) > 0$.

It remains to check incentive compatibility. The new no-inspection actions are all weakly below $\theta_e - \rho$. Hence types

$$t > \theta_e - \rho$$

cannot accept them and obtain no positive deviation payoff from them. Types

$$t \leq \theta_e - \rho$$

lie in the positive-rent region. Since the rent profile is continuous, their truthful payoff is bounded away from zero on $[\underline{\theta}, \theta_e - \rho]$. The new deviation payoff created by any report in E_0 is multiplied by ε and is uniformly bounded above on this compact set. Hence, for sufficiently small ε , all such new deviation payoffs are strictly below the existing truthful payoffs. All other deviation payoffs are unchanged, and the truthful payoffs of types in E_0 are unchanged by construction.

Thus incentive compatibility is preserved, while the Principal's expected payoff strictly increases. This contradicts optimality. Therefore deterministic full inspection cannot occur on a positive-measure subset of (θ_c, θ_e) .

Combining this with the previous paragraph gives

$$0 < i(\theta) < 1, \quad \theta \in (\theta_c, \theta_e),$$

up to null sets.

We now characterize the two on-path actions in the stochastic inspection region. Fix $\theta \in (\theta_c, \theta_e)$ with $0 < i(\theta) < 1$. Suppose first that

$$a(\theta) > a^I(\theta, \theta).$$

Consider a perturbation that decreases $a(\theta)$ by $\delta > 0$ and increases $a^I(\theta, \theta)$ by $\varepsilon > 0$, while keeping the truthful payoff of type θ fixed:

$$-(1 - i(\theta))u_a(\theta, a(\theta))\delta + i(\theta)u_a(\theta, a^I(\theta, \theta))\varepsilon = 0.$$

Thus

$$\delta = \varepsilon \frac{i(\theta)u_a(\theta, a^I(\theta, \theta))}{(1 - i(\theta))u_a(\theta, a(\theta))}.$$

Because this perturbation lowers the no-inspection action, it weakly relaxes all incentive constraints. The first-order change in the Principal's payoff is

$$\begin{aligned} dV &= -(1 - i(\theta))v_a(\theta, a(\theta))\delta + i(\theta)v_a(\theta, a^I(\theta, \theta))\varepsilon \\ &= \varepsilon i(\theta)u_a(\theta, a^I(\theta, \theta)) \left[MC(\theta, a(\theta)) - MC(\theta, a^I(\theta, \theta)) \right]. \end{aligned}$$

By Assumption 1vi, $MC(\theta, a)$ is strictly increasing in a . Since $a(\theta) > a^I(\theta, \theta)$, we have $dV > 0$, contradicting optimality. Therefore

$$a^I(\theta, \theta) \geq a(\theta).$$

Equality is impossible. Suppose

$$a^I(\theta, \theta) = a(\theta).$$

Since type θ receives strictly positive payoff, Assumption 1ii implies

$$a(\theta) = a^I(\theta, \theta) > \theta.$$

Reduce $i(\theta)$ slightly, lower $a(\theta)$ just enough so that the deviation payoff generated by report θ does not increase, and raise $a^I(\theta, \theta)$ so that the truthful payoff of type θ remains unchanged. All incentive constraints are preserved because the

no-inspection deviation payoff does not increase and detected mimics are punished. Since the two actions initially coincide, the action adjustment has only second-order effect by Assumption 1i, while the saving in inspection cost is first order. This contradicts optimality.

$$x = a(\theta) = a^I(\theta, \theta), \quad 0 < i(\theta) < 1.$$

Reduce inspection by a small amount $\eta > 0$. Set the new uninspected action to $x - d_\eta$ and the new inspected action to $x + h_\eta$. Choose d_η so that mimicking report θ does not become more attractive. For a mimicker t , the no-inspection deviation payoff changes from

$$(1 - i(\theta))u(t, x)$$

to approximately

$$(1 - i(\theta) + \eta)u(t, x - d_\eta).$$

Using Taylor expansion,

$$\begin{aligned} & (1 - i(\theta) + \eta)u(t, x - d_\eta) - (1 - i(\theta))u(t, x) \\ &= \eta u(t, x) - (1 - i(\theta))u_a(t, x)d_\eta + O(\eta d_\eta + d_\eta^2). \end{aligned}$$

So choosing $d_\eta = K\eta$ with K large enough makes this weakly negative. Thus deviation payoffs do not increase.

Now choose h_η so that type θ 's truthful payoff is unchanged:

$$(1 - i(\theta) + \eta)u(\theta, x - d_\eta) + (i(\theta) - \eta)u(\theta, x + h_\eta) = u(\theta, x).$$

Taylor expanding around x ,

$$u_a(\theta, x) \left[- (1 - i(\theta) + \eta)d_\eta + (i(\theta) - \eta)h_\eta \right] + O(d_\eta^2 + h_\eta^2) = 0.$$

Since $d_\eta = O(\eta)$, this implies

$$h_\eta = \frac{1 - i(\theta)}{i(\theta)}d_\eta + O(\eta^2),$$

so also $h_\eta = O(\eta)$.

The Principal's payoff change is

$$\Delta V = (1 - i(\theta) + \eta)v(\theta, x - d_\eta) + (i(\theta) - \eta)v(\theta, x + h_\eta) - v(\theta, x) + \eta\phi.$$

Taylor expanding,

$$\Delta V = v_a(\theta, x) \left[- (1 - i(\theta) + \eta)d_\eta + (i(\theta) - \eta)h_\eta \right] + O(d_\eta^2 + h_\eta^2) + \eta\phi.$$

But the bracket is exactly the same first-order term that appears in the promise-keeping condition, so it is only $O(d_\eta^2 + h_\eta^2)$. Therefore

$$\Delta V = \eta\phi + O(d_\eta^2 + h_\eta^2) = \eta\phi + O(\eta^2) > 0$$

for small enough η .

So the action adjustments affect the Principal only at order η^2 , while the inspection-cost saving is order η . That is why equality $a^I(\theta, \theta) = a(\theta)$ cannot be optimal.

Hence

$$a^I(\theta, \theta) > a(\theta).$$

Finally, suppose

$$a(\theta) \leq \theta.$$

Since type θ receives strictly positive payoff and $a^I(\theta, \theta) > a(\theta)$, Assumption [1ii](#) implies

$$a^I(\theta, \theta) > \theta.$$

Choose $\delta > 0$ small enough that $\theta + \delta < \theta_e$. Replace $a(\theta)$ by $\theta + \delta$, and choose $\tilde{a}^I(\theta, \theta) < a^I(\theta, \theta)$ so that the truthful payoff of type θ is unchanged:

$$(1 - i(\theta))u(\theta, \theta + \delta) + i(\theta)u(\theta, \tilde{a}^I(\theta, \theta)) = i(\theta)u(\theta, a^I(\theta, \theta)).$$

Hence

$$\tilde{a}^I(\theta, \theta) = a^I(\theta, \theta) - \frac{1 - i(\theta)}{i(\theta)} \frac{u_a(\theta, \theta)}{u_a(\theta, a^I(\theta, \theta))} \delta + o(\delta).$$

The change in the Principal's payoff is

$$\begin{aligned} \Delta V = (1 - i(\theta)) & \left[v(\theta, \theta + \delta) - v(\theta, a(\theta)) \mathbb{1}_{\{a(\theta) \geq \theta\}} \right] \\ & + i(\theta) \left[v(\theta, \tilde{a}^I(\theta, \theta)) - v(\theta, a^I(\theta, \theta)) \right]. \end{aligned}$$

If $a(\theta) < \theta$, then the first term converges to

$$(1 - i(\theta))v(\theta, \theta) > 0,$$

so $\Delta V > 0$ for all sufficiently small δ . If $a(\theta) = \theta$, then using $v_a(\theta, \theta) = 0$ and the expansion above,

$$\Delta V = (1 - i(\theta))u_a(\theta, \theta)MC(\theta, a^I(\theta, \theta))\delta + o(\delta) > 0,$$

because $a^I(\theta, \theta) > \theta$ and Assumption 1vi implies $MC(\theta, a^I(\theta, \theta)) > MC(\theta, \theta) = 0$.

The new no-inspection action is below θ_e , so types above θ_e cannot obtain positive payoff from it. Types below θ_e have positive rents, and by continuity of the rent profile incentive compatibility is preserved for sufficiently small δ . This contradicts optimality. Therefore

$$a(\theta) > \theta.$$

Thus

$$a^I(\theta, \theta) > a(\theta) > \theta, \quad \theta \in (\theta_e, \theta_e),$$

up to null sets and payoff-irrelevant boundary assignments.

No deterministic full-inspection region of positive measure.

It remains to rule out a deterministic full-inspection region of strictly positive measure outside the positive-rent region. For $\theta \geq \theta_e$, the highest payoff available from the no-inspection branches is zero. By the no-overdelivery argument above, the truthful payoff of such types is therefore zero. By Assumption 1ii, any accepted action that gives type θ zero payoff must be exactly θ . Hence any served zero-rent type must be served at the efficient action.

Suppose, toward a contradiction, that there is a deterministic full-inspection set of strictly positive measure. Then there exist $\theta_0 < \theta_1$ such that, for every $\theta \in [\theta_0, \theta_1]$,

$$i(\theta) = 1, \quad a^I(\theta, \theta) = \theta, \quad \text{the truthful payoff of type } \theta \text{ is } 0.$$

Fix $\varepsilon > 0$. For every $\theta \in [\theta_0, \theta_1]$, reduce the inspection probability from 1 to $1 - \varepsilon$ and set the no-inspection action equal to θ . The truthful payoff remains zero because

$$u(\theta, \theta) = 0$$

by Assumption 1ii. The Principal saves

$$\varepsilon \phi[F(\theta_1) - F(\theta_0)],$$

which is of order ε because $f(\theta) > 0$ on $[\underline{\theta}, \bar{\theta}]$.

This perturbation may create new deviations by lower types. For every $t < \theta_1$, the new deviation payoff from reports in $[\theta_0, \theta_1]$ is bounded above by

$$\varepsilon u(t, \theta_1).$$

By Assumptions 1i and 1ii, $u(t, \theta_1)$ is bounded on $[\underline{\theta}, \theta_1]$.

First consider zero-rent types $t \in [\theta_e, \theta_1)$. For each such t , increase the truthful accepted action from t to $t + d_\varepsilon(t)$ so that

$$u(t, t + d_\varepsilon(t)) - u(t, t) = \varepsilon u(t, \theta_1).$$

Since $u_a(t, t) > 0$ by Assumption 1ii, there is a constant $C < \infty$ such that

$$d_\varepsilon(t) \leq C\varepsilon$$

for all sufficiently small ε . By Assumption 1v,

$$v(t, t) - v(t, t + d_\varepsilon(t)) \leq K d_\varepsilon(t)^\gamma \leq K C^\gamma \varepsilon^\gamma.$$

Since $\gamma > 1$, the aggregate compensation cost for these zero-rent types is $o(\varepsilon)$.

Now consider types $t < \theta_e$. Only types whose original truthful payoff is smaller than the new deviation payoff require compensation. Let

$$B_\varepsilon = \{t < \theta_e : \text{truthful payoff of type } t < \varepsilon u(t, \theta_1)\}.$$

Since the rent profile is continuous and strictly positive below θ_e , and since $u(t, \theta_1)$ is bounded on $[\underline{\theta}, \theta_1]$, we have

$$\int_{B_\varepsilon} f(t) dt \rightarrow 0 \quad \text{as } \varepsilon \downarrow 0.$$

For each $t \in B_\varepsilon$, raise the inspection probability, if necessary, to at least ε :

$$\tilde{i}(t) = \max\{i(t), \varepsilon\}.$$

This increases the inspection probability by at most ε , so the extra inspection cost over B_ε is bounded by

$$\varepsilon \phi \int_{B_\varepsilon} f(t) dt = o(\varepsilon).$$

It also weakly lowers the probability of the no-inspection branch, and therefore does

not increase any deviation payoff.

Now increase only the truthful inspected action of each $t \in B_\varepsilon$ so that its truthful payoff is at least

$$\varepsilon u(t, \theta_1).$$

The needed increase in expected utility is of order ε for each $t \in B_\varepsilon$. Since $\tilde{i}(t) \geq \varepsilon$ and Assumption 1ii gives $u_a > 0$, this can be achieved with a bounded increase in the inspected action. Because the action space is bounded without loss by Proposition 6 and Assumption 1i holds, the Principal's cost of this compensation is bounded by

$$C\varepsilon \int_{B_\varepsilon} f(t) dt = o(\varepsilon)$$

for some finite constant C .

The compensation creates no new deviation payoff, because it is delivered only through truthful inspected branches and, under full commitment, detected mimics are punished. Hence all incentive constraints are preserved. The total compensation cost is $o(\varepsilon)$, while the inspection-cost saving on $[\theta_0, \theta_1]$ is strictly of order ε . For sufficiently small ε , the perturbation strictly increases the Principal's expected payoff, contradicting optimality. Therefore no deterministic full-inspection region can have strictly positive measure.

Proof of part (iii): Exclusion Region.

Consider types above θ_e . These types receive zero truthful payoff and cannot be served on a positive-measure set without creating a deterministic full-inspection region of positive measure, which has just been ruled out. Hence, up to null sets, all types $\theta > \theta_e$ are excluded. Exclusion is implemented by

$$i(\theta) = 0, \quad a(\theta) = \underline{\theta}.$$

For every $\theta > \theta_e$, except for payoff-irrelevant boundary cases, $\underline{\theta} < \theta$, so by Assumption 1ii the Agent rejects and both players receive zero.

Combining the preceding steps, the optimal mechanism is characterized, up to null sets and payoff-irrelevant boundary assignments, by thresholds

$$\underline{\theta} \leq \theta_c \leq \theta_e \leq \bar{\theta}.$$

For $\theta < \theta_c$, there is no inspection and

$$a(\theta) = a_c \geq \theta_c.$$

For $\theta \in (\theta_c, \theta_e)$, inspection is stochastic and

$$a^I(\theta, \theta) > a(\theta) > \theta.$$

For $\theta > \theta_e$, the Agent is excluded by the rejected action $a(\theta) = \underline{\theta}$. The equalities $\theta_c = \underline{\theta}$, $\theta_c = \theta_e$, and $\theta_e = \bar{\theta}$ are allowed, so any of the three regions may be empty. ■

Proof of Proposition 1

Proof. The proof is immediate by Proposition 7 ■

Proposition 8 Suppose the Principal has full commitment power but is restricted to deterministic inspection rules, $i(\theta) \in \{0, 1\}$, and suppose Assumption 1i, Assumption 1ii, and Assumption 1iv hold. Let θ_c^D solve

$$\max_{\theta_c \in [\underline{\theta}, \bar{\theta}]} \left\{ \int_{\underline{\theta}}^{\theta_c} v(\theta, \theta_c) dF(\theta) + \int_{\theta_c}^{\bar{\theta}} (v(\theta, \theta) - \phi) \mathbb{1}_{\{v(\theta, \theta) \geq \phi\}} dF(\theta) \right\}.$$

Given θ_c^D , define

$$\theta_e^D = \sup \{ \theta \in [\theta_c^D, \bar{\theta}] : v(\theta, \theta) \geq \phi \},$$

with the convention that $\theta_e^D = \theta_c^D$ if the set is empty. Then an optimal deterministic mechanism is given by

$$i(\theta) = \begin{cases} 0, & \theta < \theta_c^D, \\ 1, & \theta_c^D \leq \theta \leq \theta_e^D, \\ 0, & \theta > \theta_e^D, \end{cases}$$

and

$$a(\theta) = \begin{cases} \theta_c^D, & \theta < \theta_c^D, \\ \theta, & \theta_c^D \leq \theta \leq \theta_e^D, \\ \underline{\theta}, & \theta > \theta_e^D. \end{cases}$$

The truthful inspected action is efficient,

$$a^I(\theta, \theta) = \theta,$$

and every detected misreport is punished by

$$a^I(\hat{\theta}, \theta) = \underline{\theta}, \quad \hat{\theta} \neq \theta.$$

Proof. Under full commitment, detected misreports can be punished maximally. Hence, whenever $\hat{\theta} \neq \theta$, set

$$a^I(\hat{\theta}, \theta) = \underline{\theta}.$$

By Assumption [1ii](#), this gives the deviating Agent payoff zero. Therefore, if type θ mimics report $\hat{\theta}$, the only possible positive payoff comes from the no-inspection branch:

$$(1 - i(\hat{\theta}))u(\theta, a(\hat{\theta}))\mathbb{1}_{\{a(\hat{\theta}) \geq \theta\}}.$$

Thus incentive compatibility requires

$$\begin{aligned} & i(\theta)u(\theta, a^I(\theta, \theta))\mathbb{1}_{\{a^I(\theta, \theta) \geq \theta\}} + (1 - i(\theta))u(\theta, a(\theta))\mathbb{1}_{\{a(\theta) \geq \theta\}} \\ & \geq (1 - i(\hat{\theta}))u(\theta, a(\hat{\theta}))\mathbb{1}_{\{a(\hat{\theta}) \geq \theta\}}, \quad \forall \theta, \hat{\theta}. \end{aligned}$$

If $a(\theta) < \theta$, the Agent rejects the action. Hence, without loss of generality, any rejected no-inspection action can be set equal to $\underline{\theta}$. Given any deterministic mechanism, define

$$\theta_c^D = \sup\{a(\theta) : i(\theta) = 0\}.$$

By the reduced incentive constraint, taking reports with no-inspection actions arbitrarily close to θ_c^D gives

$$\begin{aligned} & i(\theta)u(\theta, a^I(\theta, \theta))\mathbb{1}_{\{a^I(\theta, \theta) \geq \theta\}} + (1 - i(\theta))u(\theta, a(\theta))\mathbb{1}_{\{a(\theta) \geq \theta\}} \\ & \geq u(\theta, \theta_c^D)\mathbb{1}_{\{\theta_c^D \geq \theta\}}. \end{aligned}$$

Consider first a type with $i(\theta) = 0$. If the no-inspection action is accepted, then $a(\theta) \geq \theta$. The previous inequality implies $a(\theta) \geq \theta_c^D$ whenever $\theta \leq \theta_c^D$, while the definition of θ_c^D implies $a(\theta) \leq \theta_c^D$. Hence

$$a(\theta) = \theta_c^D \quad \text{for accepted no-inspection types.}$$

If $\theta > \theta_c^D$, no no-inspection action can be accepted, because any accepted action would satisfy $a(\theta) \geq \theta > \theta_c^D$, contradicting the definition of θ_c^D . Therefore a

deterministic mechanism satisfying IC must have

$$a(\theta) = \begin{cases} \theta_c^D, & \theta < \theta_c^D, \\ \underline{\theta}, & \theta > \theta_c^D, \end{cases}$$

up to boundary assignments, whenever $i(\theta) = 0$.

Now consider a type with $i(\theta) = 1$. If $\theta \leq \theta_c^D$, incentive compatibility requires

$$a^I(\theta, \theta) \geq \theta_c^D.$$

If $\theta > \theta_c^D$, then any inspected type that is served must receive an accepted action, and ex-post participation requires

$$a^I(\theta, \theta) \geq \theta.$$

Therefore any inspected type must satisfy

$$a^I(\theta, \theta) \geq \max\{\theta_c^D, \theta\}.$$

By Assumption [1iv](#), the Principal's payoff is decreasing in the action on the region $a > \theta$. Therefore, conditional on inspection, the optimal inspected action is

$$a^I(\theta, \theta) = \max\{\theta_c^D, \theta\}.$$

Using the previous observations, the Principal's problem can be written as

$$\max_{i(\cdot), \theta_c^D} \int_{\underline{\theta}}^{\bar{\theta}} [i(\theta)(v(\theta, \max\{\theta_c^D, \theta\}) - \phi) + (1 - i(\theta))v(\theta, \theta_c^D)\mathbb{1}_{\{\theta_c^D \geq \theta\}}] dF(\theta).$$

If $\theta \leq \theta_c^D$, then

$$v(\theta, \max\{\theta_c^D, \theta\}) - \phi = v(\theta, \theta_c^D) - \phi < v(\theta, \theta_c^D),$$

so the optimal deterministic choice is

$$i(\theta) = 0 \quad \text{for } \theta \leq \theta_c^D.$$

If $\theta > \theta_c^D$, then the no-inspection action is rejected and gives the Principal payoff zero, while inspection gives

$$v(\theta, \theta) - \phi.$$

Hence the optimal deterministic choice is

$$i(\theta) = 1 \quad \text{if and only if} \quad v(\theta, \theta) \geq \phi.$$

Because Assumption 1iv implies that $v(\theta, \theta)$ is strictly decreasing in θ , the set of types above θ_c^D for which inspection is profitable is an interval ending at θ_e^D . Therefore the optimal deterministic objective reduces to

$$\max_{\theta_c \in [\underline{\theta}, \bar{\theta}]} \left\{ \int_{\underline{\theta}}^{\theta_c} v(\theta, \theta_c) dF(\theta) + \int_{\theta_c}^{\bar{\theta}} (v(\theta, \theta) - \phi) \mathbb{1}_{\{v(\theta, \theta) \geq \phi\}} dF(\theta) \right\}.$$

Let θ_c^D solve this problem and let θ_e^D be defined as in the statement. The candidate deterministic mechanism is therefore

$$i(\theta) = \begin{cases} 0, & \theta < \theta_c^D, \\ 1, & \theta_c^D \leq \theta \leq \theta_e^D, \\ 0, & \theta > \theta_e^D, \end{cases}$$

with

$$a(\theta) = \begin{cases} \theta_c^D, & \theta < \theta_c^D, \\ \theta, & \theta_c^D \leq \theta \leq \theta_e^D, \\ \underline{\theta}, & \theta > \theta_e^D, \end{cases}$$

and

$$a^I(\theta, \theta) = \theta, \quad a^I(\hat{\theta}, \theta) = \underline{\theta} \quad \text{for } \hat{\theta} \neq \theta.$$

It remains to verify global incentive compatibility. Types $\theta < \theta_c^D$ receive the no-inspection action θ_c^D . They are indifferent among all reports below θ_c^D , since those reports give the same action. They cannot profitably mimic reports in $[\theta_c^D, \theta_e^D]$, because those reports are inspected with probability one and detected misreports are punished. They also cannot profitably mimic reports above θ_e^D , because those reports are excluded.

Types $\theta \geq \theta_c^D$ receive zero payoff, either because they are efficiently served after inspection or because they are excluded. Such a type cannot profitably mimic a report below θ_c^D , since that report gives the no-inspection action $\theta_c^D \leq \theta$, which yields at most zero payoff by Assumption 1ii. He also cannot profitably mimic any report in $[\theta_c^D, \theta_e^D]$, because inspection is certain and detected misreports are punished. Finally, reports above θ_e^D are excluded and yield zero payoff. Thus the mechanism is globally incentive compatible.

Therefore the mechanism described in the statement is optimal among deterministic full-commitment mechanisms. ■

Proof of Proposition 2

Proof. Fix an equilibrium under commitment only to actions. We first derive an upper bound on the Principal's ex-ante payoff and then show that this upper bound is achieved by the optimal deterministic full-commitment mechanism characterized in Proposition 8.

Let $U(\theta)$ denote type θ 's equilibrium payoff. Define the no-inspection set with positive payoff by

$$\Theta_N = \{\theta : i(m(\theta)) = 0 \text{ and } U(\theta) > 0\},$$

and define the inspected set by

$$\Theta_I = \{\theta : i(m(\theta)) > 0\}.$$

By Assumption 1ii, the Agent strictly prefers higher actions. Hence all types in Θ_N must pool on one common no-inspection action. Denote this action by

$$a^* = \sup_{\{m: i(m)=0\}} a(m).$$

We first show that the no-inspection types must lie below the inspected types. Suppose, toward a contradiction, that there exist

$$\theta_I < \theta_N, \quad \theta_I \in \Theta_I, \quad \theta_N \in \Theta_N.$$

Since θ_N obtains strictly positive payoff from the no-inspection pool, we have

$$a^* > \theta_N > \theta_I.$$

Let m be the message sent by type θ_I .

Consider first the case

$$a(m) \geq a^*.$$

If $i(m) = 1$, incentive compatibility for type θ_I against mimicking the no-inspection pool implies

$$u(\theta_I, a^I(m, \theta_I)) \geq u(\theta_I, a^*).$$

By Assumption [1ii](#), this implies

$$a^I(m, \theta_I) \geq a^*.$$

Thus assigning a^* without inspection gives the Principal at least

$$v(\theta_I, a^*) \geq v(\theta_I, a^I(m, \theta_I)),$$

and strictly more than

$$v(\theta_I, a^I(m, \theta_I)) - \phi,$$

by Assumption [1iv](#) and because $\phi > 0$.

If instead $i(m) \in (0, 1)$, sequential rationality of the Principal requires her to be indifferent between inspecting and not inspecting after message m . Hence her continuation payoff after message m is equal to the payoff from the no-inspection branch, namely $v(\theta_I, a(m))$. Since $a(m) \geq a^* > \theta_I$, Assumption [1iv](#) implies

$$v(\theta_I, a^*) \geq v(\theta_I, a(m)).$$

Thus replacing the allocation of type θ_I by no inspection and action a^* weakly raises the Principal's payoff.

We must also check that type θ_I does not acquire a profitable deviation. Since type θ_N optimally takes the no-inspection pool, incentive compatibility for θ_N implies that, for every message m' ,

$$u(\theta_N, a^*) \geq (1 - i(m'))u(\theta_N, a(m')).$$

The inspected branch of any deviation can be punished after inspection, so it does not give a positive deviation payoff. If $a(m') \leq a^*$, then Assumption [1ii](#) implies

$$u(\theta_I, a^*) \geq u(\theta_I, a(m')).$$

If $a(m') > a^*$, then, since $\theta_I < \theta_N < a^* < a(m')$, Assumption [1iii](#) implies

$$\frac{u(\theta_I, a(m'))}{u(\theta_I, a^*)} \leq \frac{u(\theta_N, a(m'))}{u(\theta_N, a^*)}.$$

Combining this with the incentive constraint of θ_N gives

$$u(\theta_I, a^*) \geq (1 - i(m'))u(\theta_I, a(m')).$$

Thus type θ_I does not want to mimic any other message.

Now consider the case

$$a(m) < a^*.$$

In order for type θ_I not to mimic the no-inspection pool, incentive compatibility requires

$$i(m)u(\theta_I, a^I(m, \theta_I)) + (1 - i(m))u(\theta_I, a(m)) \geq u(\theta_I, a^*).$$

Since $a(m) < a^*$ and Assumption 1ii gives $u_a > 0$, this requires

$$a^I(m, \theta_I) > a^*.$$

Replacing the allocation of type θ_I by no inspection and action a^* therefore weakly raises the Principal's payoff, and strictly raises it whenever inspection was used with positive probability in a payoff-relevant way, because

$$v(\theta_I, a^*) > v(\theta_I, a^I(m, \theta_I)) - \phi$$

by Assumption 1iv and $\phi > 0$. The same incentive-compatibility argument from the previous case shows that type θ_I has no profitable deviation after receiving the certain action a^* .

Therefore, in any payoff-maximizing equilibrium, the no-inspection positive-rent types can be taken to lie below the inspected types, up to null sets. We write this ordering as

$$\Theta_N < \Theta_I,$$

meaning that every type in Θ_N is weakly below every type in Θ_I , up to null sets.

We now derive an upper bound on the Principal's payoff. For types in Θ_N , the Principal's payoff is

$$v(\theta, a^*)\mathbb{1}_{\{a^* \geq \theta\}}.$$

For types in Θ_I , the inspection probability satisfies $i(m) > 0$. If $i(m) = 1$, the Principal's payoff from type θ is

$$v(\theta, a^I(m, \theta)) - \phi \leq v(\theta, \theta) - \phi,$$

because, by ex-post participation and Assumption 1ii, any accepted action must be at least θ , and Assumption 1iv implies that the Principal's payoff is maximized at the efficient action θ . If $i(m) \in (0, 1)$, sequential rationality requires the Principal to be indifferent between inspecting and not inspecting after message m . Hence her

continuation payoff is equal to her inspection payoff, which is again bounded above by

$$v(\theta, \theta) - \phi.$$

Thus every equilibrium payoff is bounded above by the value of the relaxed problem

$$\mathbb{P}_a(a^*, \Theta_N, \Theta_I) = \int_{\Theta_N} v(\theta, a^*) \mathbb{1}_{\{a^* \geq \theta\}} dF(\theta) + \int_{\Theta_I} [v(\theta, \theta) - \phi] dF(\theta),$$

subject to the ordering constraint

$$\Theta_N < \Theta_I.$$

For the purpose of maximizing this upper bound, it is without loss to consider inspected types only above the no-inspection action a^* . Indeed, if an inspected type $\theta < a^*$ must be prevented from mimicking the no-inspection pool, then his inspected allocation must deliver at least the utility generated by a^* . Since inspection costs $\phi > 0$, assigning this type to the no-inspection action a^* weakly raises the Principal's payoff and does not create a new deviation by the log-supermodularity argument above. Therefore the maximum under the ordering constraint is bounded above by the relaxed problem with

$$\Theta_I \subseteq [a^*, \bar{\theta}].$$

For fixed a^* , the best such upper bound is obtained by assigning all types below a^* to the no-inspection pool and inspecting exactly those types above a^* for which verified efficient service is profitable. Therefore the upper bound becomes

$$\max_{a^* \in [\underline{\theta}, \bar{\theta}]} \left\{ \int_{\underline{\theta}}^{a^*} v(\theta, a^*) dF(\theta) + \int_{a^*}^{\bar{\theta}} [v(\theta, \theta) - \phi] \mathbb{1}_{\{v(\theta, \theta) \geq \phi\}} dF(\theta) \right\}.$$

This is exactly the deterministic full-commitment problem $\mathbb{P}_D$ characterized in Proposition 8. Hence the Principal's payoff under commitment only to actions is bounded above by the value of the optimal deterministic full-commitment mechanism.

It remains to show that this upper bound is achievable. Let θ_c^D and θ_e^D be the thresholds from Proposition 8. Consider two messages, m_N and m_I . Types

$$[\underline{\theta}, \theta_c^D) \cup (\theta_e^D, \bar{\theta}]$$

send m_N , while types

$$[\theta_c^D, \theta_e^D]$$

send m_I . The Principal commits to

$$a(m_N) = \theta_c^D, \quad a^I(m_N, \theta) = \underline{\theta} \quad \forall \theta,$$

and

$$a(m_I) = \underline{\theta},$$

with

$$a^I(m_I, \theta) = \begin{cases} \theta, & \theta \in [\theta_c^D, \theta_e^D], \\ \underline{\theta}, & \theta \notin [\theta_c^D, \theta_e^D]. \end{cases}$$

After message m_N , the Principal strictly prefers not to inspect. If she does not inspect, low types receive the pool action θ_c^D and high types reject. If she inspects, the committed inspected action is $\underline{\theta}$, which is rejected except possibly at a payoff-irrelevant boundary, and the Principal also pays the inspection cost.

After message m_I , the Principal weakly prefers to inspect. If she does not inspect, the action is $\underline{\theta}$ and the Agent rejects, giving the Principal payoff zero. If she inspects, she obtains

$$\mathbb{E}[v(\theta, \theta) - \phi \mid \theta \in [\theta_c^D, \theta_e^D]],$$

which is nonnegative because

$$v(\theta, \theta) \geq \phi \quad \text{for all } \theta \in [\theta_c^D, \theta_e^D].$$

Thus the inspection choices are sequentially rational.

Finally, the Agent's reporting incentives coincide with those in the optimal deterministic full-commitment mechanism. Types below θ_c^D accept the pooling action θ_c^D and do not want to send m_I , because after inspection they would be identified as outside $[\theta_c^D, \theta_e^D]$ and receive the punishment action $\underline{\theta}$. Types in $[\theta_c^D, \theta_e^D]$ receive zero payoff from truthful reporting and cannot profitably mimic m_N , because $\theta_c^D \leq \theta$. Types above θ_e^D are excluded and also cannot profit from sending m_I , because inspection reveals their type and triggers the punishment action $\underline{\theta}$.

Therefore the Principal can achieve exactly the payoff of the optimal deterministic full-commitment mechanism. Since this payoff is also an upper bound, the maximum payoff under commitment only to actions is equal to the optimal deterministic full-commitment payoff. ■

Proof of Lemma 1

Proof. (i) Let m_1 and m_2 be two messages on the equilibrium path that the Principal inspects with probability less than one, i.e. $i(m_1) < 1$ and $i(m_2) < 1$. Let $\Theta_i = \{\theta \in [\underline{\theta}, \bar{\theta}] : m(\theta) = m_i\}$ for $i \in \{1, 2\}$. Let

$$\theta_{m_i} = \sup \left\{ \theta | m = m_i, \text{ and } \int_{t < \theta} \beta(\theta | m = m_i) d\theta = 0 \right\}.$$

By definition, the probability measure of types less than θ_{m_i} that send message m_i is zero. By a simple computation, for a small enough $\delta > 0$

$$\begin{aligned} & \int_{\underline{\theta}}^{\theta_{m_i} + \delta} \left[v(\theta, \theta_{m_i} + \delta) \right] \beta(\theta | m = m_i) d\theta \\ &= \int_{\underline{\theta}}^{\theta_{m_i}} \left[v(\theta, \theta_{m_i} + \delta) \right] \beta(\theta | m = m_i) d\theta + \int_{\theta_{m_i}}^{\theta_{m_i} + \delta} \left[v(\theta, \theta_{m_i} + \delta) \right] \beta(\theta | m = m_i) d\theta \\ &= \int_{\theta_{m_i}}^{\theta_{m_i} + \delta} \left[v(\theta, \theta_{m_i} + \delta) \right] \beta(\theta | m = m_i) d\theta > 0. \end{aligned}$$

From second line to third line is by the fact that probability measure of types less than θ_{m_i} is zero and $v(\cdot, \cdot)$ is bounded. The above value is strictly positive since $v(\theta, \theta) > 0$ and the probability measure of types between θ_{m_i} and $\theta_{m_i} + \delta$ is strictly positive. As a consequence, for $i \in \{1, 2\}$ we have

$$\max_{\tilde{a}} \int_{\underline{\theta}}^{\tilde{a}} \left[v(\theta, \tilde{a}) \right] \beta(\theta | m = m_i) d\theta > 0.$$

Best reply of the Principal $a(m)$ is

$$a(m) = \operatorname{argmax}_{\tilde{a}} \int_{\underline{\theta}}^{\bar{\theta}} \left[V(\theta, \tilde{a}) \mathbb{1}_{\tilde{a} \geq \theta} \right] \beta(\theta | m) d\theta = \int_{\underline{\theta}}^{\tilde{a}} \left[V(\theta, \tilde{a}) \right] \beta(\theta | m) d\theta.$$

The maximum is strictly positive, therefore $a(m_i) > \inf\{\Theta_i\}$. Moreover for all $\epsilon > 0$ small enough we can find a type $\theta_i \in \Theta_i$ such that $a(m_i) - \theta_i < \epsilon$. Otherwise $a(m_i)$ is not optimal and the Principal can decrease it.

Now by contradiction assume $a(m_1) > a(m_2)$, then types in Θ_2 that are very close but less than $a(m_2)$ want to deviate and send message m_1 . Formally

$$(1 - i(m_1))u(\theta, a(m_1))\mathbb{1}_{a(m_1) \geq \theta} > (1 - i(m_2))u(\theta, a(m_2))\mathbb{1}_{a(m_2) \geq \theta},$$

for $\theta \in \Theta_2$ and small ϵ such that $\theta \in (a(m_2) - \epsilon, a(m_2))$. The reason is for small enough ϵ , the right side goes to zero, and the left side is always higher than a positive amount. A contradiction. Hence $a(m_1) = a(m_2)$.

(ii) By contradiction assume $i(m_1) > i(m_2)$. By (i) $a(m_1) = a(m_2)$. Let $\Theta_i = \{\theta \in [\underline{\theta}, \bar{\theta}] : m(\theta) = m_i\}$ for $i \in \{1, 2\}$. Then types in Θ_1 that are less than $a(m_1)$ want to deviate to m_2 . A contradiction. ■

Proof of Proposition 3

Proof. By Lemma 1, it is enough to consider equilibria with at most two on-path messages, denoted by m_N and m_I . Fix an equilibrium $\mathbf{E}$. Let

$$\Theta_I = \{\theta : m(\theta) = m_I\}, \quad \Theta_N = \{\theta : m(\theta) = m_N\},$$

and let $\mathfrak{M}(S)$ denote the probability measure of a set S .

Claim 1 In any equilibrium under commitment only to inspection,

$$\Theta_I \subseteq [a(m_N), \bar{\theta}].$$

Moreover, the equilibrium payoff is equivalent to the value of the problem

$$\begin{aligned} \mathbb{P}_I : \quad & \int_{\Theta_N} \left[(1 - i(m_N))v(\theta, a(m_N))\mathbb{1}_{\{a(m_N) \geq \theta\}} + i(m_N)(v(\theta, \theta) - \phi) \right] dF(\theta) \\ & + \int_{\Theta_I} [v(\theta, \theta) - \phi] dF(\theta), \end{aligned}$$

subject to

$$\Theta_I \subseteq [a(m_N), \bar{\theta}]$$

and

$$a(m_N) \in \operatorname{argmax}_{\tilde{a}} \int_{\Theta_N} v(\theta, \tilde{a}) \mathbb{1}_{\{\tilde{a} \geq \theta\}} dF(\theta).$$

Proof of Claim 1. If some type in Θ_I were below $a(m_N)$, then that type would strictly prefer to send m_N , because with positive probability he would receive the accepted action $a(m_N)$, whereas after inspection the Principal assigns the ex-post optimal action and gives zero rent. Hence $\Theta_I \subseteq [a(m_N), \bar{\theta}]$. The objective above is exactly the Principal's payoff from the two messages: after m_N , the Principal inspects with probability $i(m_N)$ and otherwise chooses $a(m_N)$; after m_I ,

the Principal inspects and assigns the ex-post optimal action. The last display is the Principal's sequential optimality condition after observing m_N . ■

Let

$$a(m_N) = \tilde{\theta}_c^D.$$

The value of $i(m_N)$ in $\mathbb{P}_I$ does not affect the constraints. Hence, unless full inspection is itself optimal, the Principal chooses $i(m_N) = 0$. Thus we focus on the case

$$i(m_N) = 0, \quad i(m_I) = 1.$$

We next show that the inspected set can be taken to be an interval. Fix an optimal inspected set Θ_I . Holding its probability measure fixed, move the inspected mass as far down as possible above $\tilde{\theta}_c^D$. Since $v(\theta, \theta)$ is decreasing in θ , this weakly increases the value of inspection. Moreover, this change does not alter the optimality of $a(m_N) = \tilde{\theta}_c^D$ after the no-inspection message: for actions below $\tilde{\theta}_c^D$, the payoff comparison is unchanged, and for actions above $\tilde{\theta}_c^D$, the payoff from not inspecting weakly decreases. Therefore the inspected set can be written as

$$\Theta_I = [\tilde{\theta}_c^D, \tilde{\theta}_e^D]$$

for some

$$\tilde{\theta}_c^D \leq \tilde{\theta}_e^D \leq \bar{\theta}.$$

Thus types in

$$[\underline{\theta}, \tilde{\theta}_c^D] \cup [\tilde{\theta}_e^D, \bar{\theta}]$$

send m_N , and types in $[\tilde{\theta}_c^D, \tilde{\theta}_e^D]$ send m_I . The Principal's problem can therefore be written as

$$\mathbb{P}_2 : \quad \max_{\tilde{\theta}_c^D, \tilde{\theta}_e^D} \left\{ \int_{\underline{\theta}}^{\tilde{\theta}_c^D} v(\theta, \tilde{\theta}_c^D) dF(\theta) + \int_{\tilde{\theta}_e^D}^{\tilde{\theta}_e^D} [v(\theta, \theta) - \phi] dF(\theta) \right\},$$

subject to

$$\tilde{\theta}_c^D \in \operatorname{argmax}_{\tilde{s}} \int_{\theta \in [\underline{\theta}, \tilde{\theta}_c^D] \cup [\tilde{\theta}_e^D, \bar{\theta}]} v(\theta, \tilde{s}) \mathbb{1}_{\{\tilde{s} \geq \theta\}} dF(\theta).$$

This proves the threshold structure in part (i).

We now prove part (ii). Suppose, toward a contradiction, that

$$\tilde{\theta}_e^D < \theta_e^D.$$

Increase the upper threshold from $\tilde{\theta}_e^D$ to θ_e^D . The objective in $\mathbb{P}_2$ weakly increases, because the types added to the inspected region are precisely types that are inspected in the optimal deterministic inspection policy. It remains to check that the constraint defining $\tilde{\theta}_c^D$ is unchanged. For every $\tilde{s} \leq \tilde{\theta}_c^D$, the relevant payoff comparison is unchanged up to the common rescaling of the posterior over the no-inspection message. For every $\tilde{s} \geq \theta_e^D > \tilde{\theta}_e^D$, the payoff from choosing $\tilde{s}$ after m_N weakly decreases when the interval $[\tilde{\theta}_e^D, \theta_e^D]$ is moved from m_N to m_I . Hence $\tilde{\theta}_c^D$ remains optimal after m_N . Therefore the change is feasible and weakly raises the Principal's payoff, contradicting optimality of $\tilde{\theta}_e^D$. Thus

$$\theta_e^D \leq \tilde{\theta}_e^D.$$

Finally, consider part (iii). Suppose

$$\phi < v(\bar{\theta}, \bar{\theta}),$$

so that the deterministic inspection benchmark has no exclusion region, i.e.

$$\theta_e^D = \bar{\theta}.$$

By part (ii),

$$\tilde{\theta}_e^D = \bar{\theta}.$$

It remains to identify the lower threshold. We show that

$$a(m_N) = \tilde{\theta}_c^D = \theta_c^D.$$

Suppose first that

$$a(m_N) < \theta_c^D.$$

Then

$$\int_{\underline{\theta}}^{a(m_N)} v(\theta, a(m_N)) dF(\theta) > \int_{\underline{\theta}}^{\theta_c^D} v(\theta, \theta_c^D) dF(\theta).$$

Adding the inspected payoff for the remaining types gives a payoff strictly above the value of the deterministic inspection benchmark, contradicting the optimality of θ_c^D in that benchmark. Hence $a(m_N) \not< \theta_c^D$.

Now suppose

$$a(m_N) > \theta_c^D.$$

Since after m_I the Principal puts zero posterior probability on types in the pooling

region, this would imply

$$a(m_N) > \theta_e^D.$$

But $\theta_e^D = \bar{\theta}$, which is impossible. Therefore

$$a(m_N) = \theta_e^D.$$

Thus, when the deterministic benchmark has no exclusion region, the optimal equilibrium under commitment only to inspection is payoff-equivalent to the optimal deterministic inspection policy. ■

Assumption 2 For all $(\theta, \theta', s) \in [\underline{\theta}, \bar{\theta}]^3$, such that $\theta' \leq \theta \leq s$,

$$v(\theta, s) - v(\theta, \theta) \geq v(\theta', s) - v(\theta', \theta').$$

It is easy to see that Assumption 2 is a weaker condition than supermodularity.

Proposition 9 The payoff function

$$v(\theta, a) = v_0(\theta) - (a - \theta)^\beta, \quad \beta > 1,$$

satisfies Assumption 2.

Proof. Fix any $\theta', \theta, s \in [\underline{\theta}, \bar{\theta}]$ such that $\theta' \leq \theta \leq s$. Then

$$v(\theta, s) - v(\theta, \theta) = -(s - \theta)^\beta$$

and

$$v(\theta', s) - v(\theta', \theta') = -(s - \theta')^\beta.$$

Since $\theta' \leq \theta \leq s$, we have

$$s - \theta' \geq s - \theta \geq 0.$$

Because $x \mapsto x^\beta$ is increasing on $\mathbb{R}_+$,

$$(s - \theta')^\beta \geq (s - \theta)^\beta.$$

Therefore

$$-(s - \theta)^\beta \geq -(s - \theta')^\beta,$$

which implies

$$v(\theta, s) - v(\theta, \theta) \geq v(\theta', s) - v(\theta', \theta').$$

Thus Assumption 2 holds. ■

Proof of Proposition 4

Proof. By Lemma 1 and Claim 1, it is enough to consider two on-path messages, m_N and m_I . Let

$$\Theta_I = \{\theta : m(\theta) = m_I\}, \quad \Theta_N = \{\theta : m(\theta) = m_N\}.$$

Let

$$a(m_N) = \theta_c^{NC}.$$

Claim 1 implies

$$\Theta_I \subseteq [\theta_c^{NC}, \bar{\theta}].$$

Since we restrict attention to finite-interval strategies, the only remaining step is to show that Θ_I is a single interval beginning at θ_c^{NC} .

Suppose, toward a contradiction, that Θ_I is not a single interval. Then there exist adjacent intervals $[t_1, t_2]$ and $[t_2, t_3]$ above θ_c^{NC} such that

$$m(t) = m_N \quad \text{for } t \in [t_1, t_2], \quad m(t) = m_I \quad \text{for } t \in [t_2, t_3].$$

Choose subintervals $[t'_1, t_2] \subseteq [t_1, t_2]$ and $[t_2, t'_2] \subseteq [t_2, t_3]$ with the same probability measure under $F(\cdot)$. Switch their messages: types in $[t'_1, t_2]$ now send m_I , and types in $[t_2, t'_2]$ now send m_N . Let β_s denote the posterior after this switch.

First, the switch does not change the optimal no-inspection action after m_N . For every $\tilde{a} \leq \theta_c^{NC}$, the switched types are all above θ_c^{NC} and therefore do not accept $\tilde{a}$. Hence the payoff comparison that made θ_c^{NC} optimal is unchanged. For every $\tilde{a} \geq \theta_c^{NC}$, the no-inspection payoff from choosing $\tilde{a}$ weakly decreases, because the no-inspection message replaces lower types by higher types of the same measure. Therefore θ_c^{NC} remains optimal after m_N . Moreover, the value of inspecting after m_N strictly decreases, because $v(\theta, \theta)$ is decreasing in θ . Hence the Principal still does not inspect after m_N .

It remains to show that the Principal still inspects after m_I . Let $a_s(m_I)$ be the best no-inspection action after m_I following the switch. Because the switched intervals can be chosen arbitrarily small, an optimal $a_s(m_I)$ cannot lie inside (t_1, t'_2) : if it did, lowering the action to t_1 would keep the same relevant accepting set and strictly raise the Principal's payoff.

If $a_s(m_I) < t_1$, then the value of not inspecting after m_I is unchanged by the switch, while the value of inspecting after m_I strictly increases because the inspection

message replaces higher types by lower types of the same measure. Hence inspection remains optimal after m_I .

Now suppose $a_s(m_I) > t'_2$. The increase in the value of inspecting after m_I is

$$\frac{1}{\mathfrak{M}(\Theta_I)} \left[\int_{t'_1}^{t_2} v(\theta, \theta) dF(\theta) - \int_{t_2}^{t'_2} v(\theta, \theta) dF(\theta) \right].$$

The increase in the value of not inspecting after m_I is at most

$$\frac{1}{\mathfrak{M}(\Theta_I)} \left[\int_{t'_1}^{t_2} v(\theta, a_s(m_I)) dF(\theta) - \int_{t_2}^{t'_2} v(\theta, a_s(m_I)) dF(\theta) \right].$$

By Assumption 2, the first expression is weakly larger than the second. Therefore, if inspection was optimal after m_I before the switch, it remains optimal after the switch.

Thus the switch preserves sequential optimality after both messages. It also weakly increases the Principal's ex-ante payoff, and strictly increases it whenever the inspected and no-inspected intervals are non-null, because lower types are moved into the inspected message and higher types are moved into the no-inspection message. This contradicts optimality of an equilibrium in which Θ_I is not connected. Therefore

$$\Theta_I = [\theta_c^{NC}, \theta_e^{NC}]$$

for some

$$\theta_c^{NC} \leq \theta_e^{NC} \leq \bar{\theta}.$$

Consequently, types in

$$[\underline{\theta}, \theta_c^{NC}] \cup [\theta_e^{NC}, \bar{\theta}]$$

send m_N , while types in $[\theta_c^{NC}, \theta_e^{NC}]$ send m_I . After m_N , the Principal does not inspect and mandates $a(m_N) = \theta_c^{NC}$. After m_I , she inspects and, because she has no commitment power, mandates the ex-post optimal action $a^I(m_I, \theta) = \theta$. This proves the proposition. ■

Proof of Proposition 5

Proof. Since

$$\phi < v(\bar{\theta}, \bar{\theta}),$$

and since Assumption [1iv](#) implies that $v(\theta, \theta)$ is decreasing in θ , we have

$$v(\theta, \theta) - \phi > 0 \quad \forall \theta \in [\underline{\theta}, \bar{\theta}].$$

Hence, in the optimal deterministic full-commitment mechanism characterized in Proposition [8](#), the exclusion threshold is

$$\theta_e^D = \bar{\theta}.$$

Let θ_c^D denote the corresponding deterministic no-inspection threshold.

We construct an equilibrium that implements the same payoff. There are two messages, m_N and m_I . The equilibrium reporting strategy is

$$m(\theta) = \begin{cases} m_N, & \theta \in [\underline{\theta}, \theta_c^D], \\ m_I, & \theta \in (\theta_c^D, \bar{\theta}]. \end{cases}$$

The Principal is committed to inspect after message m_I . After message m_N , she chooses whether to inspect sequentially. If she does not inspect after m_N , she mandates the action $a(m_N)$. If she inspects, she observes the type and mandates the efficient action. Thus

$$a^I(m, \theta) = \theta \quad \forall m, \theta.$$

Off path, beliefs put probability one on the lowest type:

$$\beta(\underline{\theta} \mid m) = 1, \quad m \notin \{m_N, m_I\}.$$

We first show that, after observing m_N , the Principal optimally chooses

$$a(m_N) = \theta_c^D$$

if she does not inspect. Given the equilibrium strategy, the posterior after m_N is supported on $[\underline{\theta}, \theta_c^D]$. Hence the Principal's no-inspection action solves

$$a(m_N) \in \arg \max_{\tilde{a} \in [\underline{\theta}, \bar{\theta}]} \int_{\underline{\theta}}^{\min\{\tilde{a}, \theta_c^D\}} v(\theta, \tilde{a}) dF(\theta),$$

where types above $\tilde{a}$ reject the action. If $\tilde{a} > \theta_c^D$, all types in the posterior accept, but Assumption [1iv](#) implies that lowering $\tilde{a}$ to θ_c^D weakly raises the Principal's payoff. Thus no maximizer lies above θ_c^D .

Now suppose, toward a contradiction, that some $\tilde{a} < \theta_c^D$ gives a strictly higher no-inspection payoff than θ_c^D :

$$\int_{\underline{\theta}}^{\tilde{a}} v(\theta, \tilde{a}) dF(\theta) > \int_{\underline{\theta}}^{\theta_c^D} v(\theta, \theta_c^D) dF(\theta).$$

Then the deterministic full-commitment objective evaluated at threshold $\tilde{a}$ would be strictly larger than the objective evaluated at θ_c^D , because all types above $\tilde{a}$ can be inspected profitably:

$$\begin{aligned} & \int_{\underline{\theta}}^{\tilde{a}} v(\theta, \tilde{a}) dF(\theta) + \int_{\tilde{a}}^{\bar{\theta}} [v(\theta, \theta) - \phi] dF(\theta) \\ & > \int_{\underline{\theta}}^{\theta_c^D} v(\theta, \theta_c^D) dF(\theta) + \int_{\theta_c^D}^{\bar{\theta}} [v(\theta, \theta) - \phi] dF(\theta). \end{aligned}$$

This contradicts the definition of θ_c^D in Proposition 8. Therefore $a(m_N) = \theta_c^D$ is optimal after message m_N whenever the Principal does not inspect.

Next, we show that the Principal does not want to inspect after observing m_N . If she does not inspect, her payoff is

$$\int_{\underline{\theta}}^{\theta_c^D} v(\theta, \theta_c^D) dF(\theta).$$

If she inspects, her payoff is

$$\int_{\underline{\theta}}^{\theta_c^D} [v(\theta, \theta) - \phi] dF(\theta).$$

Suppose the latter were strictly larger. Then full inspection of all types would yield a strictly higher deterministic full-commitment payoff than the threshold θ_c^D :

$$\begin{aligned} & \int_{\underline{\theta}}^{\theta_c^D} [v(\theta, \theta) - \phi] dF(\theta) + \int_{\theta_c^D}^{\bar{\theta}} [v(\theta, \theta) - \phi] dF(\theta) \\ & > \int_{\underline{\theta}}^{\theta_c^D} v(\theta, \theta_c^D) dF(\theta) + \int_{\theta_c^D}^{\bar{\theta}} [v(\theta, \theta) - \phi] dF(\theta), \end{aligned}$$

again contradicting the optimality of θ_c^D in Proposition 8. Thus the Principal weakly prefers not to inspect after m_N , and we select no inspection after m_N .

We now check the Agent's incentives. A type $\theta \leq \theta_c^D$ obtains

$$u(\theta, \theta_c^D) \geq 0$$

from sending m_N . If he sends m_I , then the Principal inspects and assigns the efficient action θ , giving payoff

$$u(\theta, \theta) = 0.$$

Thus he does not profitably deviate to m_I . Off-path messages yield the action $\underline{\theta}$ under the specified beliefs and hence do not give a strictly positive payoff.

A type $\theta > \theta_c^D$ sends m_I , is inspected, and receives the efficient action θ , so his payoff is

$$u(\theta, \theta) = 0.$$

If he deviates to m_N , then he receives the no-inspection action $\theta_c^D < \theta$, which he rejects by Assumption [1ii](#). Hence the deviation gives payoff zero. Off-path messages also give no positive payoff. Thus no type has a profitable deviation.

Finally, after any off-path message $m \notin \{m_N, m_I\}$, the belief assigns probability one to $\underline{\theta}$. The Principal then optimally does not inspect and assigns $\underline{\theta}$, since inspection only adds the cost ϕ and the efficient action for the believed type is $\underline{\theta}$.

The equilibrium payoff is therefore

$$\int_{\underline{\theta}}^{\theta_c^D} v(\theta, \theta_c^D) dF(\theta) + \int_{\theta_c^D}^{\bar{\theta}} [v(\theta, \theta) - \phi] dF(\theta),$$

which is exactly the payoff of the optimal deterministic full-commitment mechanism in Proposition [8](#). Hence the Principal can attain the optimal deterministic full-commitment payoff under partial commitment. ■

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